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Chapter 1

Electromagnetic wave : Basics

1.1 Summary of Electromagnetic laws

1. Here are both differential and integral forms of Maxwell's equations:

TABLE 9.1 Generalized Forms of Maxwell's Equations

Differential Form	Integral Form	Remarks
$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of isolated magnetic charge*
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int_S \mathbf{B} \cdot d\mathbf{S}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$	Ampère's circuit law

*This is also referred to as Gauss's law for magnetic fields.

2. Continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

is implicit in Maxwell's equations. By taking the divergence of Ampere's circuit law and using the Gauss's law, we can derive the continuity equation as follows:

$$\begin{aligned}
 \nabla \cdot \nabla \times \mathbf{H} &= \nabla \cdot \mathbf{J} + \frac{\partial(\nabla \cdot \mathbf{D})}{\partial t} \\
 &= \nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} \\
 &= 0 \quad (\because \text{divergence of curl is zero})
 \end{aligned}$$

3. The Lorentz force equation

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

4. The constitutive relations in a linear, homogeneous, and isotropic medium are characterized by the conductivity(σ), permittivity(ϵ), and permeability(μ).

$$\begin{aligned}
 \mathbf{D} &= \epsilon \mathbf{E} = \epsilon_0 \mathbf{E} + \mathbf{P} \\
 \mathbf{B} &= \mu \mathbf{H} = \mu_0 (\mathbf{H} + \mathbf{M}) \\
 \mathbf{J} &= \sigma \mathbf{E} + \rho_v \mathbf{v}
 \end{aligned}$$

5. Since the constitutive relations hold for time-varying electromagnetic fields, the electromagnetic boundary conditions also remain valid.

$$\begin{aligned}
 E_{1t} - E_{2t} &= 0 \quad \text{or} \quad (\mathbf{E}_1 - \mathbf{E}_2) \times \hat{n} = 0 \\
 H_{1t} - H_{2t} &= K \quad \text{or} \quad (\mathbf{H}_1 - \mathbf{H}_2) \times \hat{n} = \mathbf{K} \\
 D_{1n} - D_{2n} &= \rho_s \quad \text{or} \quad (\mathbf{D}_1 - \mathbf{D}_2) \cdot \hat{n} = \rho_s \\
 B_{1n} - B_{2n} &= 0 \quad \text{or} \quad (\mathbf{B}_2 - \mathbf{B}_1) \cdot \hat{n} = 0
 \end{aligned}$$

Here, $\hat{n}$ denotes the unit normal vector to the boundary.

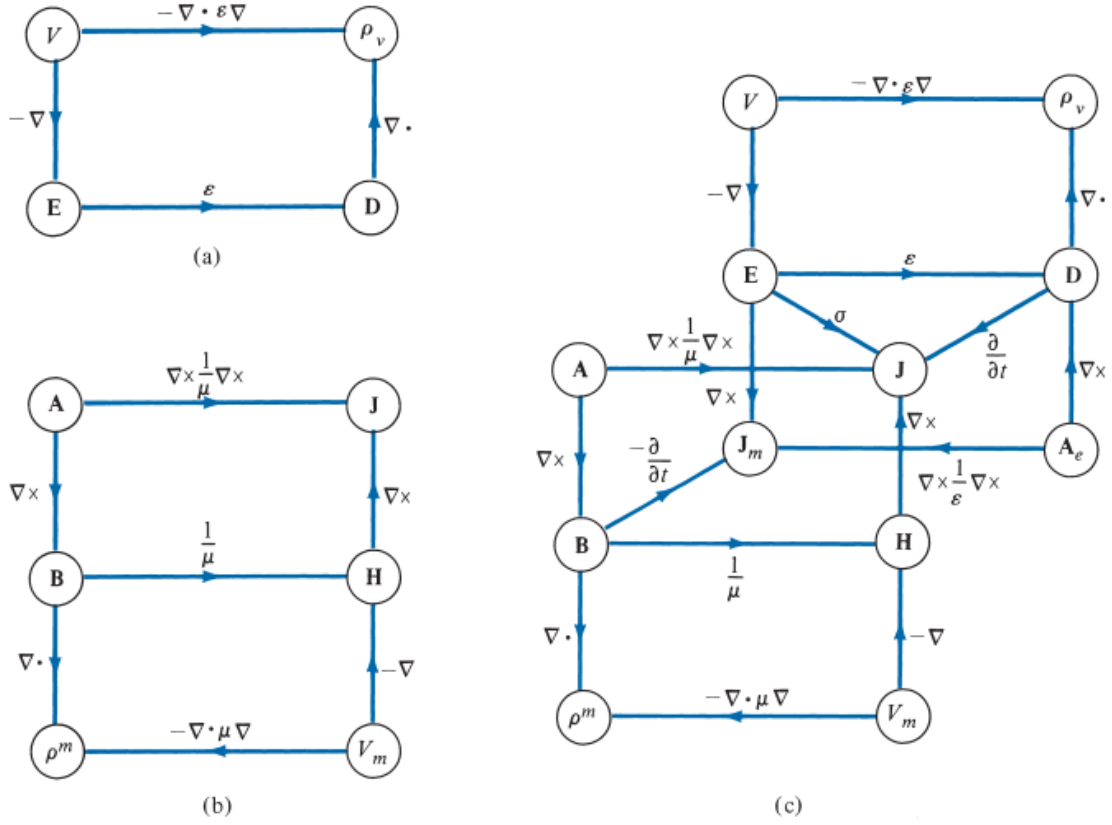


FIGURE 9.11 Electromagnetic flow diagrams showing the relationship between the potentials and vector fields: (a) electrostatic system, (b) magnetostatic system, (c) electromagnetic system. [Adapted with permission from the Publishing Department of the Institution of Electrical Engineers.]

1. The above figure is the electromagnetic flow diagram that helps with visualization of the basic relationships between field quantities.
2. In the electrostatic system (See Figure 9.11a, here ρ_v denotes the volume charge density),
 - (a) Given the potential V , compute the electric field $\mathbf{E}$: $\mathbf{E} = -\nabla V$
 - (b) Given $\mathbf{E}$, compute the displacement field $\mathbf{D}$: $\mathbf{D} = \epsilon \mathbf{E}$
 - (c) Given $\mathbf{D}$, compute ρ_v : $\rho_v = \nabla \cdot \mathbf{D}$
 - (d) Given V , compute ρ_v : $\rho_v = \nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon \mathbf{E}) = \nabla \cdot (-\epsilon \nabla V) = (-\nabla \cdot \epsilon \nabla) V$
3. In the magnetostatic system (See Figure. 9.11b, here μ denotes the permeability of the medium),
 - (a) Given the potential $\mathbf{A}$, compute the magnetic flux density $\mathbf{B}$: $\mathbf{B} = \nabla \times \mathbf{A}$
 - (b) Given the magnetic flux density $\mathbf{B}$, compute the magnetic field intensity $\mathbf{H}$:

$$\mathbf{H} = \frac{1}{\mu} \mathbf{B}$$
 - (c) Given $\mathbf{H}$, compute $\mathbf{J}$: $\mathbf{J} = \nabla \times \mathbf{H}$
 - (d) Given $\mathbf{A}$, compute $\mathbf{J}$: $\mathbf{J} = \nabla \times \mathbf{H} = \nabla \times \left(\frac{\mathbf{B}}{\mu} \right) = \nabla \times \left(\frac{1}{\mu} \nabla \times \mathbf{A} \right)$

1.2 Maxwell's Equations : Time-harmonic fields and phasor forms

1. If a vector $\mathbf{A}(\mathbf{r}, t)$ (instantaneous form) is a time-harmonic field, the **phasor** form of $\mathbf{A}$ is $\mathbf{A}_s$ (here, $\mathbf{r} = (x, y, z)$); Two quantities are related as

$$\mathbf{A}(\mathbf{r}, t) = \text{Re} [\mathbf{A}_s(\mathbf{r})e^{j\omega t}]$$

Note that the phasor is a function of position, not a function of time.

2. **Converting the instantaneous form to the phasor form.** For example, if $\mathbf{A} = A_o \cos(kx - \omega t)\hat{y}$, we can write $\mathbf{A}$ as

$$\mathbf{A} = \text{Re} [A_o e^{jkx}\hat{y} \cdot e^{-j\omega t}]$$

Thus, the phasor form of $\mathbf{A}$ is

$$\mathbf{A}_s = A_o e^{jkx}\hat{y}$$

If we take the time derivative,

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{\partial}{\partial t} \text{Re}(\mathbf{A}_s e^{-j\omega t}) = \text{Re}(-j\omega \mathbf{A}_s e^{-j\omega t})$$

Thus, the time derivative of the instantaneous quantity is equivalent to multiplying its phasor form by $(-j\omega)$. That is,

$$\frac{\partial \mathbf{A}}{\partial t} \rightarrow (-j\omega) \mathbf{A}_s$$

Similarly,

$$\int \mathbf{A} \partial t \rightarrow \frac{\mathbf{A}_s}{(-j\omega)}$$

3. Example 1.

Given that

$$\mathbf{A} = 10 \cos(10^8 t - 10x + 60^\circ) \mathbf{a}_z, \quad \mathbf{B}_s = \frac{20}{j} \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y,$$

express $\mathbf{A}$ in phasor form and $\mathbf{B}_s$ in instantaneous form.

Solution:

$$\begin{aligned}\mathbf{A} &= \text{Re}[10e^{j(\omega t - 10x + 60^\circ)} \mathbf{a}_z], \quad \omega = 10^8, \\ &= \text{Re}[10e^{j(60^\circ - 10x)} \mathbf{a}_z e^{j\omega t}] = \text{Re}[\mathbf{A}_s e^{j\omega t}], \\ \mathbf{A}_s &= 10e^{j(60^\circ - 10x)} \mathbf{a}_z.\end{aligned}$$

Now for $\mathbf{B}_s$:

$$\begin{aligned}\mathbf{B}_s &= \frac{20}{j} \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y = -j20 \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y \\ &= 20e^{-j\pi/2} \mathbf{a}_x + 10e^{j2\pi x/3} \mathbf{a}_y.\end{aligned}$$

The instantaneous field is

$$\begin{aligned}\mathbf{B} &= \text{Re}(\mathbf{B}_s e^{j\omega t}) \\ &= \text{Re}[20e^{j(\omega t - \pi/2)} \mathbf{a}_x + 10e^{j(\omega t + 2\pi x/3)} \mathbf{a}_y] \\ &= 20 \cos(\omega t - \pi/2) \mathbf{a}_x + 10 \cos\left(\omega t + \frac{2\pi x}{3}\right) \mathbf{a}_y \\ &= 20 \sin(\omega t) \mathbf{a}_x + 10 \cos\left(\omega t + \frac{2\pi x}{3}\right) \mathbf{a}_y.\end{aligned}$$

4. Now we derive the phasor form of Maxwell's equations. For example, we start with the instantaneous form of Faraday's law,

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{B}(\mathbf{r}, t)$$

We let $\mathbf{E}(\mathbf{r}, t) = \text{Re} [\mathbf{E}_s e^{-j\omega t}]$ and $\mathbf{B}(\mathbf{r}, t) = \text{Re} [\mathbf{B}_s e^{-j\omega t}]$. Then

$$\nabla \times [\text{Re} (\mathbf{E}_s e^{-j\omega t})] = -\frac{\partial}{\partial t} \text{Re} [\mathbf{B}_s e^{-j\omega t}]$$

Note that the curl operation operates only on $\mathbf{r} = (x, y, z)$ and $\mathbf{B}_s$ is independent of time:

$$\begin{aligned} \nabla \times \text{Re} [\mathbf{E}_s e^{-j\omega t}] &= \text{Re} [(\nabla \times \mathbf{E}_s) e^{-j\omega t}] \\ -\frac{\partial}{\partial t} \text{Re} [\mathbf{B}_s e^{-j\omega t}] &= -\text{Re} [(-j\omega) \mathbf{B}_s e^{-j\omega t}] \end{aligned}$$

Thus, the phasor form of Faraday's law is

$$\nabla \times \mathbf{E}_s = (j\omega) \mathbf{B}_s$$

5. In this manner, we derive the phasor forms of all the Maxwell's equations. Note that the time factor $e^{-j\omega t}$ factors out, resulting in the time-independent equations. Herein lies the justification for using phasors : the time factor can be suppressed in our analysis of time-harmonic fields and inserted when necessary.

- (a) Gauss's law

$$\nabla \cdot \mathbf{D}_s = \rho_{vs}$$

Here, ρ_{vs} is also the phasor form of the instantaneous charge $\rho_v(t)$.

- (b) Nonexistence of isolated magnetic charge

$$\nabla \cdot \mathbf{B}_s = 0$$

- (c) Faraday's law

$$\nabla \times \mathbf{E}_s = (j\omega) \mathbf{B}_s$$

- (d) Ampere's circuit law

$$\nabla \times \mathbf{H}_s = \mathbf{J}_s - j\omega \mathbf{D}_s$$

Note that we have assumed the time factor $e^{-j\omega t}$. It is equally possible to have assumed the time factor $e^{j\omega t}$, in which case we would need to replace every j in the above equations with $-j$.

6. Example 2.

The electric field and the magnetic field in free space are given by

$$\mathbf{E} = \frac{50}{\rho} \cos(10^6 t + \beta z) \mathbf{a}_\phi \text{ V/m}, \quad \mathbf{H} = \frac{H_0}{\rho} \cos(10^6 t + \beta z) \mathbf{a}_\rho \text{ A/m}.$$

Express these in phasor form and determine H_0 and β such that the fields satisfy Maxwell's equations.

Solution:

$$\begin{aligned}\mathbf{E} &= \text{Re}(\mathbf{E}_s e^{j\omega t}), & \mathbf{H} &= \text{Re}(\mathbf{H}_s e^{j\omega t}), & \omega &= 10^6, \\ \mathbf{E}_s &= \frac{50}{\rho} e^{j\beta z} \mathbf{a}_\phi, & \mathbf{H}_s &= \frac{H_0}{\rho} e^{j\beta z} \mathbf{a}_\rho.\end{aligned}\quad (9.7.2)$$

Note that $\mathbf{a}_\phi, \mathbf{a}_\rho$ are the unit vectors in the cylindrical coordinate. For free space ($\rho_v = \sigma = 0, \varepsilon = \varepsilon_0, \mu = \mu_0$), Maxwell's equations become

$$\nabla \cdot \mathbf{E}_s = 0, \quad (9.7.3)$$

$$\nabla \cdot \mathbf{H}_s = 0, \quad (9.7.4)$$

$$\nabla \times \mathbf{H}_s = j\omega\varepsilon_0 \mathbf{E}_s, \quad (9.7.5)$$

$$\nabla \times \mathbf{E}_s = -j\omega\mu_0 \mathbf{H}_s. \quad (9.7.6)$$

Substituting (9.7.2) into (9.7.3)–(9.7.4):

$$\begin{aligned}\nabla \cdot \mathbf{E}_s &= \frac{1}{\rho} \frac{\partial}{\partial \phi} (E_{\phi s}) = 0, \\ \nabla \cdot \mathbf{H}_s &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_{\rho s}) = 0.\end{aligned}$$

Now

$$\nabla \times \mathbf{H}_s = \nabla \times \left(\frac{H_0}{\rho} e^{j\beta z} \mathbf{a}_\rho \right) = \frac{jH_0\beta}{\rho} e^{j\beta z} \mathbf{a}_\phi. \quad (9.7.7)$$

Substitute into (9.7.5):

$$\begin{aligned}\frac{jH_0\beta}{\rho} e^{j\beta z} \mathbf{a}_\phi &= j\omega\varepsilon_0 \frac{50}{\rho} e^{j\beta z} \mathbf{a}_\phi, \\ H_0\beta &= 50\omega\varepsilon_0.\end{aligned}\quad (9.7.8)$$

Similarly, from (9.7.6):

$$\begin{aligned}-j\beta \frac{50}{\rho} e^{j\beta z} \mathbf{a}_\rho &= -j\omega\mu_0 \frac{H_0}{\rho} e^{j\beta z} \mathbf{a}_\rho, \\ \frac{H_0}{\beta} &= \frac{50}{\omega\mu_0}.\end{aligned}\quad (9.7.9)$$

Multiplying (9.7.8) and (9.7.9):

$$H_0^2 = (50)^2 \frac{\varepsilon_0}{\mu_0}, \quad H_0 = \pm 50 \sqrt{\frac{\varepsilon_0}{\mu_0}} = \pm \frac{50}{120\pi} = \pm 0.1326.$$

Dividing (9.7.8) by (9.7.9):

$$\begin{aligned}\beta^2 &= \omega^2 \mu_0 \varepsilon_0, \\ \beta &= \pm \omega \sqrt{\mu_0 \varepsilon_0} = \pm \frac{\omega}{c} = \pm \frac{10^6}{3 \times 10^8} = \pm 3.33 \times 10^{-3}.\end{aligned}$$

Final result:

$$H_0 = \pm 0.1326, \quad \beta = \pm 3.33 \times 10^{-3}.$$

Only pairs $(H_0, \beta) = (0.1326, 3.33 \times 10^{-3})$ or $(-0.1326, -3.33 \times 10^{-3})$ satisfy Maxwell's equations.

7. Example 3.

In a medium characterized by $\sigma = 0$, $\mu = \mu_0$, $\varepsilon = 4\varepsilon_0$, the field is

$$\mathbf{E} = 20 \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m.}$$

Find β and $\mathbf{H}$.

Solution (time domain):

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\partial E_y}{\partial y} = 0, \\ \nabla \times \mathbf{E} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial_x & \partial_y & \partial_z \\ 0 & E_y & 0 \end{vmatrix} = -\frac{\partial E_y}{\partial z} \mathbf{a}_x = 20\beta \cos(10^8 t - \beta z) \mathbf{a}_x. \end{aligned}$$

From Faraday's law

$$\begin{aligned} \nabla \times \mathbf{E} &= -\mu \frac{\partial \mathbf{H}}{\partial t}, \\ \mathbf{H} &= -\frac{1}{\mu} \int (\nabla \times \mathbf{E}) dt = -\frac{20\beta}{\mu} \int \cos(10^8 t - \beta z) dt \mathbf{a}_x \\ &= -\frac{20\beta}{\mu 10^8} \sin(10^8 t - \beta z) \mathbf{a}_x. \end{aligned} \tag{9.8.1}$$

Next, Ampère's law:

$$\begin{aligned} \nabla \times \mathbf{H} &= \varepsilon \frac{\partial \mathbf{E}}{\partial t}, \\ \nabla \times \mathbf{H} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial_x & \partial_y & \partial_z \\ H_x & 0 & 0 \end{vmatrix} = \frac{\partial H_x}{\partial z} \mathbf{a}_y = \frac{20\beta^2}{\mu 10^8} \cos(10^8 t - \beta z) \mathbf{a}_y. \end{aligned}$$

So

$$\mathbf{E} = \frac{1}{\varepsilon} \int (\nabla \times \mathbf{H}) dt = \frac{20\beta^2}{\mu \varepsilon 10^{16}} \sin(10^8 t - \beta z) \mathbf{a}_y.$$

Comparing with given $\mathbf{E} = 20 \sin(10^8 t - \beta z) \mathbf{a}_y$:

$$\frac{20\beta^2}{\mu \varepsilon 10^{16}} = 20 \Rightarrow \beta^2 = \omega^2 \mu \varepsilon.$$

Thus

$$\beta = \pm \omega \sqrt{\mu_0 \cdot 4\varepsilon_0} = \pm \frac{\omega}{c/2} = \pm \frac{10^8}{1.5 \times 10^8} = \pm \frac{2}{3}.$$

For isotropic medium, $\beta = \frac{2}{3}$. Substituting in (9.8.1):

$$\begin{aligned} \mathbf{H} &= -\frac{20(2/3)}{\mu_0 10^8} \sin\left(10^8 t - \frac{2}{3} z\right) \mathbf{a}_x, \\ &= \frac{1}{3\pi} \sin\left(10^8 t - \frac{2}{3} z\right) \mathbf{a}_x \text{ A/m.} \end{aligned}$$

Method 2 (using phasors):

$$\mathbf{E} = \text{Im}(\mathbf{E}_s e^{j\omega t}), \quad \mathbf{E}_s = 20e^{-j\beta z} \mathbf{a}_y, \quad \omega = 10^8.$$

Again,

$$\begin{aligned} \nabla \cdot \mathbf{E}_s &= \frac{\partial E_{ys}}{\partial y} = 0, \\ \nabla \times \mathbf{E}_s &= -j\omega\mu\mathbf{H}_s \Rightarrow \mathbf{H}_s = \frac{\nabla \times \mathbf{E}_s}{-j\omega\mu}. \\ \mathbf{H}_s &= \frac{1}{-j\omega\mu} \left[-\frac{\partial E_{ys}}{\partial z} \mathbf{a}_x \right] = \frac{20\beta}{\omega\mu} e^{-j\beta z} \mathbf{a}_x. \end{aligned} \quad (9.8.4)$$

Clearly, $\nabla \cdot \mathbf{H}_s = 0$.

From Ampère's law:

$$\nabla \times \mathbf{H}_s = j\omega\varepsilon\mathbf{E}_s \Rightarrow \mathbf{E}_s = \frac{\nabla \times \mathbf{H}_s}{j\omega\varepsilon}. \quad (9.8.5)$$

Substituting (9.8.4) into (9.8.5):

$$\mathbf{E}_s = \frac{1}{j\omega\varepsilon} \frac{\partial H_{xs}}{\partial z} \mathbf{a}_y = \frac{20\beta^2}{\omega^2\mu\varepsilon} e^{-j\beta z} \mathbf{a}_y.$$

Comparing with given $\mathbf{E}_s = 20e^{-j\beta z} \mathbf{a}_y$:

$$20 = \frac{20\beta^2}{\omega^2\mu\varepsilon} \Rightarrow \beta = \pm\omega\sqrt{\mu\varepsilon} = \pm\frac{2}{3}.$$

Thus from (9.8.4):

$$\mathbf{H}_s = \frac{20(2/3)}{10^8(4\pi \times 10^{-7})} e^{-j\beta z} \mathbf{a}_x = \frac{1}{3\pi} e^{-j\beta z} \mathbf{a}_x.$$

Therefore the instantaneous field:

$$\mathbf{H} = \text{Im}(\mathbf{H}_s e^{j\omega t}) = \pm \frac{1}{3\pi} \sin(10^8 t - \beta z) \mathbf{a}_x \text{ A/m}.$$

—

Note: since E is given as sine, we used

$$\mathbf{A} = \text{Im}(\mathbf{A}_s e^{j\omega t}).$$

If it were cosine, we would use

$$\mathbf{A} = \text{Re}(\mathbf{A}_s e^{j\omega t}).$$

For example,

$$E = 20 \cos(10^8 t - \beta z - 90^\circ) \mathbf{a}_y = \text{Re}(\mathbf{E}_s e^{j\omega t}), \quad \mathbf{E}_s = 20e^{-j\beta z - j90^\circ} \mathbf{a}_y = -j20e^{-j\beta z} \mathbf{a}_y.$$

1.3 Solving Electromagnetic Wave Equations

1.3.1 Discovery of EM Wave equations from Maxwell equations

Consider a linear, isotropic, homogeneous, lossy dielectric medium that is charge free (macroscopic $\rho_v = 0$). Assuming and suppressing the time factor $e^{j\omega t}$, the phasor forms of Maxwell's equations become

$$\begin{aligned}\nabla \cdot \mathbf{D}_s &= \varepsilon \nabla \cdot \mathbf{E}_s = 0 \rightarrow \nabla \cdot \mathbf{E}_s = 0 \\ \nabla \cdot \mathbf{B}_s &= \mu \nabla \cdot \mathbf{H}_s = 0 \rightarrow \nabla \cdot \mathbf{H}_s = 0 \\ \nabla \times \mathbf{E}_s &= -j\omega \mathbf{B}_s = -j\omega \mu \mathbf{H}_s \\ \nabla \times \mathbf{H}_s &= \mathbf{J}_s + j\omega \mathbf{D}_s = \sigma \mathbf{E}_s + j\omega \mu \mathbf{E}_s\end{aligned}$$

Taking the curl of Faraday's law gives

$$\nabla \times \nabla \times \mathbf{E}_s = -j\omega \mu (\nabla \times \mathbf{H}_s) = -j\omega \mu (\sigma + j\omega \mu) \mathbf{E}_s$$

Applying the vector identity,

$$\nabla \times \nabla \times \mathbf{E}_s = \nabla(\nabla \cdot \mathbf{E}_s) - \nabla^2 \mathbf{E}_s = -\nabla^2 \mathbf{E}_s$$

We obtain the wave equation for the phasor of the electric field:

$$\nabla^2 \mathbf{E}_s = j\omega \mu (\sigma + j\omega \mu) \mathbf{E}_s$$

By a similar procedure,

$$\begin{aligned}\nabla \times \nabla \times \mathbf{H}_s &= (\sigma + j\omega \mu)(\nabla \times \mathbf{E}_s) = (\sigma + j\omega \mu)(-j\omega \mu) \mathbf{H}_s \\ \nabla \times \nabla \times \mathbf{H}_s &= \nabla(\nabla \cdot \mathbf{H}_s) - \nabla^2 \mathbf{H}_s = -\nabla^2 \mathbf{H}_s\end{aligned}$$

the wave equation for phasor of the magnetic field intensity is obtained.

$$\nabla^2 \mathbf{H}_s = j\omega \mu (\sigma + j\omega \mu) \mathbf{H}_s$$

1.3.2 Solving the EM wave equation : The plane wave solution

1. Now the electric field and magnetic fields are completely decoupled:

$$\begin{aligned}\nabla^2 \mathbf{E}_s &= j\omega\mu(\sigma + j\omega\epsilon)\mathbf{E}_s \\ \nabla^2 \mathbf{H}_s &= j\omega\mu(\sigma + j\omega\epsilon)\mathbf{H}_s\end{aligned}$$

These two vector wave equations are known as homogeneous vector **Helmholtz's equations**. We define **the propagation constant** of the medium by

$$\gamma^2 = j\omega\mu(\sigma + j\omega\epsilon)$$

Without loss of generality, if we assume that a wave propagate along $+\mathbf{a}_z$ and that $\mathbf{E}_s$ has only an x -component,

$$\mathbf{E}_s = E_x(z)\mathbf{a}_x$$

then

$$\begin{aligned}\nabla^2 \mathbf{E}_s &= j\omega\mu(\sigma + j\omega\epsilon)\mathbf{E}_s \\ \nabla^2 E_x \mathbf{a}_x + \nabla^2 E_y \mathbf{a}_y + \nabla^2 E_z \mathbf{a}_z &= \gamma^2 (E_x \mathbf{a}_x + E_y \mathbf{a}_y + E_z \mathbf{a}_z) \\ (\nabla^2 - \gamma^2) E_x(z) &= 0 \quad (\because E_y = E_z = 0)\end{aligned}$$

In the Cartesian coordinate system,

$$\begin{aligned}(\nabla^2 - \gamma^2) E_x(z) &= 0 \\ \frac{\partial^2}{\partial x^2} E_x(z) + \frac{\partial^2}{\partial y^2} E_x(z) + \frac{\partial^2}{\partial z^2} E_x(z) - \gamma^2 E_x(z) &= 0 \\ \frac{\partial^2}{\partial z^2} E_x(z) - \gamma^2 E_x(z) &= 0\end{aligned}$$

This is a scalar wave equation, a linear homogeneous differential equation, with a solution

$$E_x(z) = A e^{-\gamma z} + B e^{\gamma z}$$

This is the plane wave solution for the EM wave equation.

2. Let's revisit the definition of the propagation constant,

$$\gamma^2 = j\omega\mu(\sigma + j\omega\epsilon) \quad (1)$$

Since the propagation constant is complex, let us denote the real and imaginary parts of the propagation constant as follows:

$$\gamma = \alpha + j\beta \quad (2)$$

Squaring the above expression (2) and comparing it with (1),

$$\begin{aligned}\gamma^2 &= \alpha^2 - \beta^2 + j(2\alpha\beta) \\ &= -\omega^2\mu\epsilon + j(\omega\mu\sigma) \\ \therefore \alpha^2 - \beta^2 &= -\omega^2\mu\epsilon\end{aligned} \quad (3)$$

Meanwhile, complex square gives

$$|\gamma^2| = \alpha^2 + \beta^2 = \sqrt{(-\omega^2\mu\varepsilon)^2 + (-\omega\mu\sigma)^2} = \omega\mu\sqrt{\sigma^2 + \omega^2\varepsilon^2} \quad (4)$$

From equations (3) and (4), we obtain

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1 \right]}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1 \right]}$$

3. Now let us revisit the plane wave solution for the scalar wave equation for the electric field,

$$E_x(z) = A e^{-\gamma z} + B e^{\gamma z}$$

Using the expression (2), we can rewrite as

$$E_x(z) = A e^{-(\alpha+j\beta)z} + B e^{(\alpha+j\beta)z}$$

Since $\alpha > 0, \beta > 0$, the field component $B e^{(\alpha+j\beta)z}$ diverges to infinity as this wave travels along the $+z$ direction. Since the field must be finite at infinity, we kill this term by setting $B = 0$ and are left with

$$E_x(z) = A e^{-(\alpha+j\beta)z}$$

With this phasor expression, we can easily obtain the instantaneous field expression,

$$\mathbf{E} = E_x(z, t) \mathbf{a}_x = \text{Re} [A e^{-\alpha z} e^{-j\beta z} e^{j\omega t}] \mathbf{a}_x = A e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x$$

which can be depicted as the following figure:

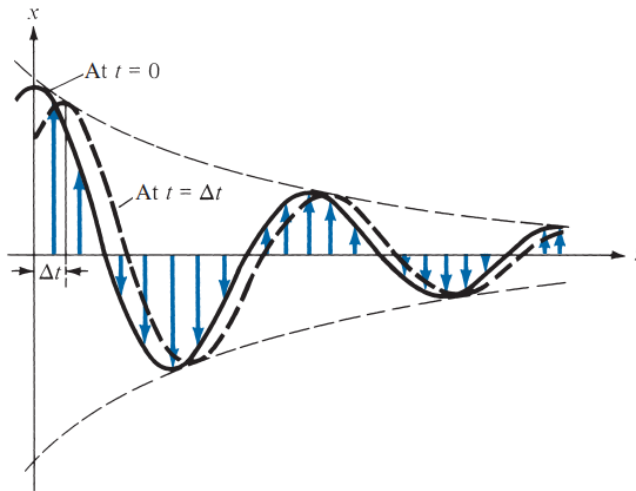


FIGURE 10.5 An E -field with an x -component traveling in the $+z$ -direction at times $t = 0$ and $t = \Delta t$; arrows indicate instantaneous values of E .

4. The magnetic field intensity can be obtained by the Ampere's circuit law,

$$\mathbf{H}_s = \frac{\nabla \times \mathbf{E}_s}{-j\omega\mu}$$

Substituting $\mathbf{E}_s = Ae^{-\gamma z}\mathbf{a}_x$ into the above equation,

$$\mathbf{H}_s = \frac{\nabla \times \mathbf{E}_s}{-j\omega\mu} = \frac{1}{-j\omega\mu} \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial_x & \partial_y & \partial_z \\ Ae^{-\gamma z} & 0 & 0 \end{vmatrix} = \frac{1}{-j\omega\mu} \frac{\partial(Ae^{-\gamma z})}{\partial z} \mathbf{a}_y = \frac{-\gamma}{-j\omega\mu} (Ae^{-\gamma z}) \mathbf{a}_y$$

Thus,

$$\mathbf{H}_s = \frac{E_x(z)}{\eta} (\mathbf{a}_z \times \mathbf{a}_x) = \frac{1}{\eta} (\mathbf{a}_z \times \mathbf{E}_s)$$

where the complex quantity η , known as the **intrinsic impedance**, in ohms, of the medium is defined as

$$\eta \equiv \frac{j\omega\mu}{\gamma} = \frac{j\omega\mu}{\sqrt{(j\omega\mu)(\sigma + j\omega\varepsilon)}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \equiv |\eta| \angle \theta_\eta$$

The intrinsic impedance accounts for the field intensity ratio $\mathbf{E}_s$ and $\mathbf{H}_s$. Skipping the algebra,

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2\right]^{1/4}}, \quad \tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}$$

where $0 \leq \theta_\eta \leq 45^\circ$. The instantaneous form of magnetic field can be written as

$$\mathbf{H} = \text{Re} \left[\frac{A}{|\eta|e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} \mathbf{a}_y \right] = \frac{A}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_y$$

1.4 Wave Polarization

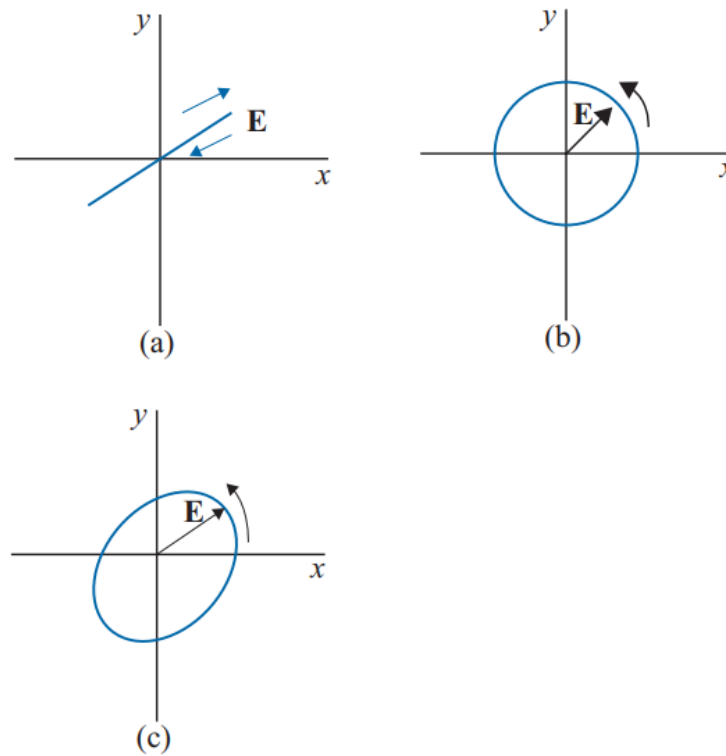


FIGURE 10.10 Wave polarizations: (a) linear, (b) circular, (c) elliptical.

1. **Polarization** may be regarded as the locus of the tip of the electric field (in a plane perpendicular to the direction of propagation) at a given point as a function of time.
2. There are three types of polarization: linear or plane, circular, and elliptical. That means that the tip of the electric field can describe a straight line, a circle, or an ellipse with time, as shown in Figure 10.10.
3. **Linear Polarization** : A uniform plane wave is linearly polarized if it has only one component or when its transverse components are in phase. For a wave traveling in the $+z$ -direction, we may have

$$E_x = E_{0x} \cos(\omega t - \beta z + \phi_x) \quad (1.1)$$

$$E_y = E_{0y} \cos(\omega t - \beta z + \phi_y) \quad (1.2)$$

where E_{0x} and E_{0y} are real. The composite wave

$$\mathbf{E} = E_{0x} \cos(\omega t - \beta z + \phi_x) \mathbf{a}_x + E_{0y} \cos(\omega t - \beta z + \phi_y) \mathbf{a}_y \quad (1.3)$$

is linearly polarized when the phase difference $\Delta\phi$ is

$$\Delta\phi = \phi_y - \phi_x = n\pi, \quad n = 0, 1, 2, \dots \quad (1.4)$$

4. **Circular polarization** : Circular polarization takes place when the x - and y -components are the same in magnitude ($E_{0x} = E_{0y} = E_0$) and the phase difference between them is an odd multiple of $\pi/2$; that is,

$$\Delta\phi = \phi_y - \phi_x = \pm(2n + 1)\pi/2, \quad n = 0, 1, 2, \dots \quad (1.5)$$

For example, the x - and y -components may be of the form

$$E_x = E_0 \cos(\omega t - \beta z) \quad (1.6)$$

$$E_y = E_0 \cos(\omega t - \beta z + \pi/2) \quad (1.7)$$

At $z = 0$, this becomes

$$E_x = E_0 \cos(\omega t), \quad E_y = E_0 \cos(\omega t + \pi/2) = -E_0 \sin(\omega t) \quad (1.8)$$

$$|E|^2 = E_x^2 + E_y^2 = E_0^2 \quad (1.9)$$

which is the equation of a circle.

5. **Elliptic polarization** : Elliptical polarization occurs when $E_{0x} \neq E_{0y}$ and the phase difference is an odd multiple of $\pi/2$:

$$\Delta\phi = \phi_y - \phi_x = \pm(2n + 1)\pi/2, \quad n = 0, 1, 2, \dots \quad (10.65)$$

For instance, at $z = 0$ with $\Delta\phi = \pi/2$, eq. (10.58) gives

$$E_x = E_{0x} \cos(\omega t) \quad \implies \quad \cos(\omega t) = \frac{E_x}{E_{0x}} \quad (1.10)$$

$$E_y = E_{0y} \cos(\omega t + \pi/2) = -E_{0y} \sin(\omega t) \quad \implies \quad -\sin(\omega t) = \frac{E_y}{E_{0y}} \quad (1.11)$$

Squaring and adding,

$$\cos^2(\omega t) + \sin^2(\omega t) = 1 \quad \implies \quad \frac{E_x^2}{E_{0x}^2} + \frac{E_y^2}{E_{0y}^2} = 1 \quad (1.12)$$

which is the equation of an ellipse. Circular and linear polarization are therefore special cases of elliptical polarization.

1.5 Power and and the Poynting vector

1.5.1 Derivation of Poynting theorem

- (a) Energy can be transported from one point (where a transmitter is located) to another point (where a receiver is located) by means of EM waves. The rate of such energy transportation can be obtained from Maxwell's equations:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1)$$

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad (2)$$

- (b) Dotting both sides of eq. (2) with $\mathbf{E}$ gives

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \sigma E^2 + \mathbf{E} \cdot \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad (3)$$

But for any vector fields $\mathbf{A}$ and $\mathbf{B}$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

Applying this vector identity to eq. (3) (letting $\mathbf{A} = \mathbf{H}$ and $\mathbf{B} = \mathbf{E}$) gives

$$\begin{aligned} \mathbf{H} \cdot (\nabla \times \mathbf{E}) + \nabla \cdot (\mathbf{H} \times \mathbf{E}) &= \mathbf{E} \cdot (\nabla \times \mathbf{H}) \\ &= \sigma E^2 + \varepsilon \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \\ &= \sigma E^2 + \frac{\varepsilon}{2} \frac{\partial}{\partial t} E^2 \end{aligned} \quad (1.13) \quad (4)$$

Meanwhile, dotting both sides of eq. (1) with $\mathbf{H}$, we write

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = \mathbf{H} \cdot \left(-\mu \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{\mu}{2} \frac{\partial}{\partial t} H^2 \quad (5)$$

and thus eq. (4) becomes

$$-\frac{\mu}{2} \frac{\partial H^2}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \sigma E^2 + \frac{\varepsilon}{2} \frac{\partial E^2}{\partial t} \quad (6)$$

- (c) Rearranging terms and taking the volume integral of both sides of (6),

$$\int_V \nabla \cdot (\mathbf{E} \times \mathbf{H}) dv = -\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 \right] dv - \int_V \sigma E^2 dv \quad (7)$$

Applying the divergence theorem to the left-hand side gives

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 \right] dv - \int_V \sigma E^2 dv \quad (8)$$

Equation (8) is referred to as **Poynting's theorem**.

1.5.2 Interpretation of Poynting theorem

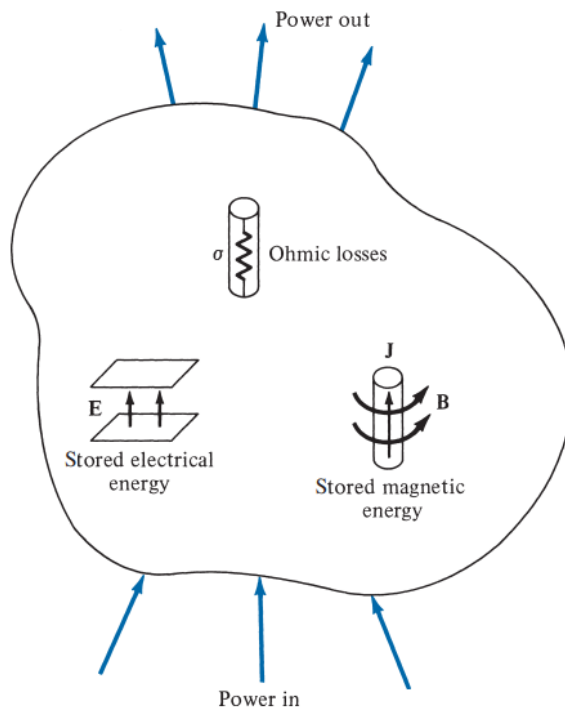


FIGURE 10.11 Illustration of power balance for EM fields.

Now let us identify the various terms in the Poynting theorem using energy-conservation arguments for EM fields.

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right] dv - \int_V \sigma E^2 dv \quad (8)$$

- (a) $\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right] dv$: The first term on the right-hand side of eq. (8) is interpreted as the rate of decrease in energy stored in the electric and magnetic fields.
- (b) $\int_V \sigma E^2 dv$: The second term is the power dissipated because the medium is conducting ($\sigma \neq 0$).
- (c) $\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S}$: The quantity $\mathbf{E} \times \mathbf{H}$ on the left-hand side of eq. (8) is known as the **Poynting vector** $\mathcal{P}$, measured in watts per square meter (W/m^2); that is,

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} \quad (9)$$

It represents the instantaneous power density vector associated with the EM field at a given point. The integration of the Poynting vector over any closed surface gives the net power flowing out of that surface.

Poynting's theorem states that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within v minus the ohmic losses.

1.5.3 Average Poynting vector

1. **Direction of Poynting vector.** It should be noted that $\mathcal{P}$ is normal to both $\mathbf{E}$ and $\mathbf{H}$ and is therefore along the direction of wave propagation $\mathbf{a}_k$ for uniform plane waves. Thus

$$\mathbf{a}_k = \mathbf{a}_E \times \mathbf{a}_H \quad (1.14)$$

The fact that $\mathcal{P}$ points along $\mathbf{a}_k$ causes $\mathcal{P}$ to be regarded as a “pointing” vector.

2. **Time-averaging of Poynting vector.** If we assume that

$$\mathbf{E}(z, t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x$$

then

$$\mathbf{H}(z, t) = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_y$$

and

$$\begin{aligned} \mathcal{P}(z, t) &= \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_z \\ &= \frac{E_0^2}{2|\eta|} e^{-2\alpha z} [\cos \theta_\eta + \cos(2\omega t - 2\beta z - \theta_\eta)] \mathbf{a}_z \end{aligned} \quad (1)$$

Since $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, to determine the time-average Poynting vector $\mathcal{P}_{\text{ave}}(z)$ (in W/m²), we integrate eq. (1) over the period $T = 2\pi/\omega$; that is,

$$\mathcal{P}_{\text{ave}}(z) = \frac{1}{T} \int_0^T \mathcal{P}(z, t) dt = \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \cos \theta_\eta \mathbf{a}_z \quad (1.15)$$

3. This time-averaged Poynting vector can easily be obtained by the phasor method. Noting that

$$\begin{aligned} \mathbf{E}_s &= E_0 e^{-(\alpha + j\beta)z} \mathbf{a}_x \\ \mathbf{H}_s &= \frac{E_0}{|\eta|} e^{-(\alpha + j\beta)z - j\theta_\eta} \mathbf{a}_y \end{aligned}$$

Then

$$\begin{aligned} \mathbf{E}_s \times \mathbf{H}_s^* &= [E_0 e^{-(\alpha + j\beta)z} \mathbf{a}_x] \times \left[\frac{E_0}{|\eta|} e^{+[(\alpha + j\beta)z + j\theta_\eta]} \mathbf{a}_y \right] \\ &= \frac{E_0^2}{|\eta|} e^{j\theta_\eta} \mathbf{a}_z \end{aligned}$$

Thus the power-average Poynting vector is equivalent to

$$\mathcal{P}_{\text{ave}}(z) = \frac{1}{2} \text{Re} (\mathbf{E}_s \times \mathbf{H}_s^*)$$

4. The total time-average power crossing a given surface S is given by

$$P_{\text{ave}} = \int_S \mathcal{P}_{\text{ave}} \cdot d\mathbf{S}$$

We should note the difference between $\mathcal{P}$, $\mathcal{P}_{\text{ave}}$, and P_{ave} : whereas $\mathcal{P}(x, y, z, t)$ is the Poynting vector in watts per square meter and is time varying, $\mathcal{P}_{\text{ave}}(x, y, z)$, also in watts per square meter, is the time average of the Poynting vector $\mathcal{P}$; it is a vector but is time invariant. Finally, P_{ave} is a total time-average power through a surface in watts; it is a scalar.

5. **Example.**

In a nonmagnetic medium

$$\mathbf{E} = 4 \sin(2\pi \times 10^7 t - 0.8x) \mathbf{a}_z \quad \text{V/m}$$

Find

- (a) ε_r , η
- (b) The time-average power carried by the wave
- (c) The total power crossing 100 cm^2 of plane $2x + y = 5$

Solution:

- (a) Since $\alpha = 0$ and $\beta \neq \omega/c$, the medium is not free space but a lossless medium:

$$\beta = 0.8, \quad \omega = 2\pi \times 10^7, \quad \mu = \mu_0 \text{ (nonmagnetic)}, \quad \varepsilon = \varepsilon_0 \varepsilon_r$$

Hence

$$\beta = \omega \sqrt{\mu \varepsilon} = \omega \sqrt{\mu_0 \varepsilon_0 \varepsilon_r} = \frac{\omega}{c} \sqrt{\varepsilon_r}$$

or

$$\sqrt{\varepsilon_r} = \frac{\beta c}{\omega} = \frac{0.8(3 \times 10^8)}{2\pi \times 10^7} = \frac{12}{\pi}$$

$$\varepsilon_r = 14.59$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_0 \varepsilon_r}} = \frac{120\pi}{\sqrt{\varepsilon_r}} = \frac{120\pi}{12/\pi} = 10\pi^2$$

$$\eta = 98.7 \Omega$$

(b)

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \frac{E_0^2}{\eta} \sin^2(\omega t - \beta x) \mathbf{a}_x$$

$$\mathcal{P}_{\text{ave}} = \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{E_0^2}{2\eta} \mathbf{a}_x = \frac{16}{2 \times 10\pi^2} \mathbf{a}_x$$

$$= 81 \mathbf{a}_x \text{ mW/m}^2$$

(c) On plane $2x + y = 5$,

$$\mathbf{a}_n = \frac{2\mathbf{a}_x + \mathbf{a}_y}{\sqrt{5}}$$

Hence the total power is

$$P_{\text{ave}} = \int_S \mathcal{P}_{\text{ave}} \cdot d\mathbf{S} = \mathcal{P}_{\text{ave}} \cdot \mathbf{S} \mathbf{a}_n$$

$$= (81 \times 10^{-3} \mathbf{a}_x)(100 \times 10^{-4}) \left[\frac{2\mathbf{a}_x + \mathbf{a}_y}{\sqrt{5}} \right]$$

$$= \frac{162 \times 10^{-5}}{\sqrt{5}}$$

$$= 724.5 \mu\text{W}$$

Chapter 2

Electromagnetic wave propagation

2.1 Plane wave propagation in unbounded medium of various properties

Now let us assemble the general results of EM plane wave propagation. Assuming the plane wave propagates in the $+z$ direction, the plane wave in general **lossy** dielectric ($\sigma > 0$) are characterized as follows:

1. The electric field of the plane wave

$$\mathbf{E} = E_x(z, t)\mathbf{a}_x = \text{Re} \left[E_0 e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right] \mathbf{a}_x = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x$$

Here, the attenuation constant α and the propagation constant β are given by

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon} \right)^2} - 1 \right]}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon} \right)^2} + 1 \right]}$$

2. The magnetic field of the plane wave

$$\mathbf{H} = H_y(z, t)\mathbf{a}_y = \text{Re} \left[\frac{E_0}{|\eta| e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} \right] \mathbf{a}_y = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_y$$

where the incident impedance η is given by

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \equiv |\eta| \angle \theta_\eta$$

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon} \right)^2 \right]^{1/4}}, \quad \tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}$$

It is important to note that $\mathbf{E}$ and $\mathbf{H}$ are out of phase by θ_η at any instant of time due to the complex intrinsic impedance of the medium. Thus, at any time, $\mathbf{E}$ leads $\mathbf{H}$ by θ_η .

3. Given the value of material parameters σ, ε, μ and the wave frequency ω , it is convenient to compute the incident impedance η first, so as to compute $\mathbf{H}$ with the given $\mathbf{E}$ or vice versa.

(a) First, compute η as follows:

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \equiv |\eta| \angle \theta_\eta$$

- (b) Given $\mathbf{E}$, compute $\mathbf{H}$: First, find the phasor expression for the electric field, $\mathbf{E}_s$. Then, use the following relationship to find $\mathbf{H}_s$:

$$\mathbf{H}_s = \frac{1}{\eta} (\mathbf{a}_z \times \mathbf{E}_s)$$

and then switch the resulting $\mathbf{H}_s$ to the instantaneous expression, $\mathbf{H}$, by inserting time factor $e^{j\omega t}$.

- (c) Given $\mathbf{H}$, compute $\mathbf{E}$: First, find the phasor expression for the magnetic field, $\mathbf{H}_s$. Then, use the following relationship to find $\mathbf{E}_s$:

$$\mathbf{E}_s = \eta (\mathbf{H}_s \times \mathbf{a}_z)$$

and then switch the resulting $\mathbf{E}_s$ to the instantaneous expression, $\mathbf{E}$, by inserting time factor $e^{j\omega t}$.

2.1.1 Loss tangent : Characterization of the medium loss

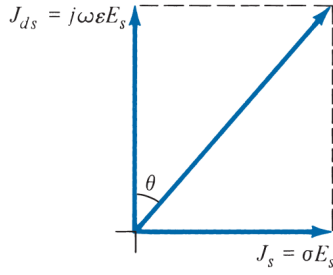


FIGURE 10.6 Loss angle of a lossy medium.

1. We characterize the degree of loss in a medium as the ratio of the magnitude of the conduction current density J_c to that of the displacement current density J_d ,

$$\frac{|J_{cs}|}{|J_{ds}|} = \frac{|\sigma E_s|}{|j\omega\epsilon E_s|} = \frac{\sigma}{\omega\epsilon} = \tan \theta$$

where $\tan \theta$ is known as the *loss tangent* and θ is the *loss angle* of the medium as illustrated in Figure 10.6.

2. A medium is said to be a good (lossless or perfect) dielectric if $\tan \theta$ is very small ($\sigma \ll \omega\epsilon$) or a good conductor if $\tan \theta$ is very large ($\sigma \gg \omega\epsilon$).
3. From the viewpoint of wave propagation, the characteristic behavior of a medium depends not only on its **constitutive parameters** σ, ϵ , and μ but also on the **frequency** of operation. A medium that is regarded as a good conductor at low frequencies may be a good dielectric at high frequencies.
4. Recall from the expression of the incident impedance

$$|\eta| = \frac{\sqrt{\mu/\epsilon}}{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2\right]^{1/4}}, \quad \tan 2\theta_\eta = \frac{\sigma}{\omega\epsilon}$$

Thus, the angle of loss tangent is two times of phase shift,

$$\theta = 2\theta_\eta$$

5. From the Ampere's circuit law, we can define the complex permittivity ϵ_c by

$$\nabla \times \mathbf{H}_s = (\sigma + j\omega\epsilon)\mathbf{E}_s = j\omega\epsilon \left[1 - \frac{j\sigma}{\omega\epsilon}\right] \mathbf{E}_s = j\omega\epsilon_c \mathbf{E}_s$$

where

$$\epsilon_c = \epsilon \left[1 - j\frac{\sigma}{\omega\epsilon}\right] = \epsilon[1 - j \tan \theta]$$

We can rewrite the complex permittivity as

$$\epsilon_c = \epsilon' - j\epsilon''$$

with $\epsilon' = \epsilon$, $\epsilon'' = \sigma/\omega$, $\epsilon = \epsilon_0\epsilon_r$.

We observe that the ratio of ϵ'' to ϵ' is the loss tangent of the medium; that is,

$$\tan \theta = \frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon}$$

2.1.2 Case 1. Plane monochromatic wave in the free space

In free space,

$$\sigma = 0, \quad \varepsilon = \varepsilon_0 = \frac{1}{36\pi} \times 10^{-9} [F/m], \quad \mu = \mu_0 = 4\pi \times 10^{-7} [H/m]$$

Then, the incident impedance of free space is

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\omega\mu_0}{0 + j\omega\varepsilon_0}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{\frac{1}{36\pi} \times 10^{-9}}} = 120\pi \approx 377 \Omega$$

so $|\eta| = 377 \Omega$ and $\theta_\eta = 0^\circ$. The attenuation and propagation constants are

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1 \right]} = \omega \sqrt{\frac{\mu_0\varepsilon_0}{2} \left[\sqrt{1 + \left(\frac{0}{\omega\varepsilon_0}\right)^2} - 1 \right]} = 0$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1 \right]} = \omega \sqrt{\frac{\mu_0\varepsilon_0}{2} \left[\sqrt{1 + \left(\frac{0}{\omega\varepsilon_0}\right)^2} + 1 \right]} = \omega \sqrt{\mu_0\varepsilon_0} = \frac{\omega}{c}$$

Hence, the field expressions are

$$\mathbf{E} = E_x(z, t)\mathbf{a}_x = \text{Re} [E_0 e^{-\alpha z} e^{-j\beta z} e^{j\omega t}] \mathbf{a}_x = E_0 \cos(\omega t - \beta z)\mathbf{a}_x$$

$$\mathbf{H} = H_y(z, t)\mathbf{a}_y = \text{Re} \left[\frac{E_0}{|\eta| e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} \right] \mathbf{a}_y = \frac{E_0}{377} \cos(\omega t - \beta z)\mathbf{a}_y$$

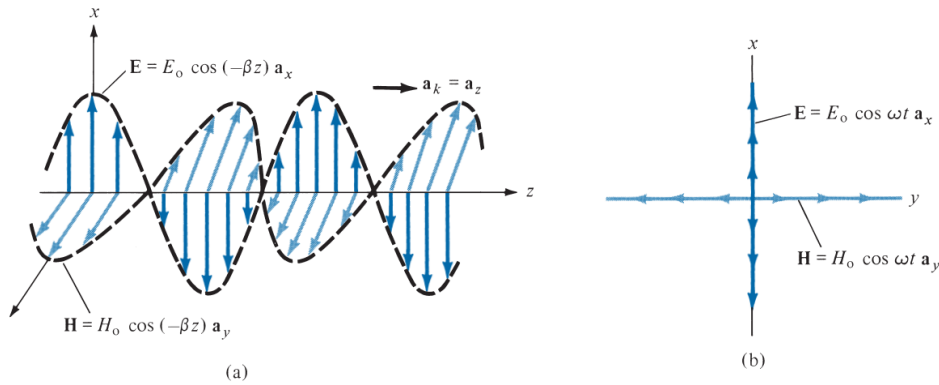


FIGURE 10.7 Plots of $\mathbf{E}$ and $\mathbf{H}$ (a) as functions of z at $t = 0$; and (b) at $z = 0$. The arrows indicate instantaneous values.

1. In general, if $\mathbf{a}_E$, $\mathbf{a}_H$, and $\mathbf{a}_k$ are unit vectors along the $\mathbf{E}$ field, the $\mathbf{H}$ field, and the direction of wave propagation

$$\mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_H$$

or

$$\mathbf{a}_k \times \mathbf{a}_H = -\mathbf{a}_E$$

or

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k$$

2. Both $\mathbf{E}$ and $\mathbf{H}$ fields (or EM waves) are everywhere normal to the direction of wave propagation, $\mathbf{a}_k$. That means that the fields lie in a plane that is transverse or orthogonal to the direction of wave propagation. They form an EM wave that has no electric or magnetic field components along the direction of propagation; such a wave is called a *transverse electromagnetic (TEM)* wave.
3. A combination of $\mathbf{E}$ and $\mathbf{H}$ is called a *uniform plane wave* because $\mathbf{E}$ (or $\mathbf{H}$) has the same magnitude throughout any transverse plane defined by $z = \text{constant}$.
4. The direction in which the electric field points is the *polarization* of a TEM wave. The wave in Figure 10.7, for example, is polarized in the x -direction.
5. A uniform plane wave cannot exist physically because it stretches to infinity and would represent infinite energy. Such waves are characteristically simple and fundamentally important. They serve as approximations to practical waves, such as those from a radio antenna at distances sufficiently far from radiating sources. Although our discussion so far deals with free space, it also applies to any other isotropic medium.

2.1.3 Case 2. Plane monochromatic wave in lossless dielectrics

In a lossless dielectric, $\sigma \ll \omega\varepsilon$,

$$\sigma \approx 0, \quad \varepsilon = \varepsilon_r \varepsilon_0, \quad \mu = \mu_r \mu_0$$

Then, the incident impedance of the lossless dielectric is

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\omega\mu}{0 + j\omega\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}}$$

so $|\eta| = \sqrt{\frac{\mu}{\varepsilon}}$ and $\theta_\eta = 0^\circ$. Thus, $\mathbf{E}$ and $\mathbf{H}$ are in time phase with each other. The attenuation and propagation constants are

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1 \right]} = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{0}{\omega\varepsilon}\right)^2} - 1 \right]} = 0$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1 \right]} = \omega \sqrt{\frac{\mu\varepsilon}{2} \left[\sqrt{1 + \left(\frac{0}{\omega\varepsilon}\right)^2} + 1 \right]} = \omega \sqrt{\mu\varepsilon} = \frac{\omega}{v}$$

Hence, the field expressions are

$$\mathbf{E} = E_x(z, t) \mathbf{a}_x = \text{Re} \left[E_0 e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right] \mathbf{a}_x = E_0 \cos(\omega t - \beta z) \mathbf{a}_x$$

$$\mathbf{H} = H_y(z, t) \mathbf{a}_y = \text{Re} \left[\frac{E_0}{|\eta| e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} \right] \mathbf{a}_y = \frac{E_0}{|\eta|} \cos(\omega t - \beta z) \mathbf{a}_y$$

2.1.4 Case 3. Plane monochromatic wave in the good conductors

In a good conductor in which $\sigma \gg \omega\epsilon$,

$$\sigma = \infty, \quad \epsilon = \epsilon_0 = \frac{1}{36\pi} \times 10^{-9} [F/m], \quad \mu = \mu_r \mu_0$$

The incident impedance is approximated as

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} \approx \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ$$

Thus $\mathbf{E}$ leads $\mathbf{H}$ by 45° . Next, the attenuation constant and propagation constant are approximated as

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]} \approx \omega \sqrt{\frac{\mu\epsilon_0}{2} \left(\frac{\sigma}{\omega\epsilon_0}\right)} = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]} \approx \omega \sqrt{\frac{\mu\epsilon_0}{2} \left(\frac{\sigma}{\omega\epsilon_0}\right)} = \sqrt{\frac{\omega\mu\sigma}{2}} = \alpha$$

Hence, if

$$\mathbf{E} = E_x(z, t) \mathbf{a}_x = \text{Re} [E_0 e^{-\alpha z} e^{-j\beta z} e^{j\omega t}] \mathbf{a}_x = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x$$

then

$$\mathbf{H} = H_y(z, t) \mathbf{a}_y = \text{Re} \left[\frac{E_0}{|\eta| e^{j\theta_\eta}} e^{-\alpha z} e^{j(\omega t - \beta z)} \right] \mathbf{a}_y = \frac{E_0}{\sqrt{\frac{\omega\mu}{\sigma}}} e^{-\alpha z} \cos(\omega t - \beta z - 45^\circ) \mathbf{a}_y$$

Skin effect

1. In the **conducting** medium, the amplitude of $\mathbf{E}$ (or $\mathbf{H}$) is attenuated by the factor $e^{-\alpha z}$. The distance δ , shown in Figure 10.8, through which the wave amplitude

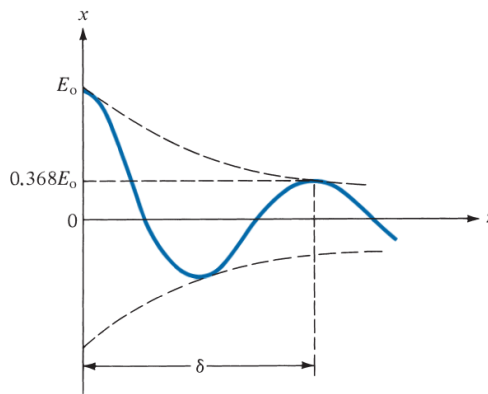


FIGURE 10.8 Illustration of skin depth.

decreases to a factor e^{-1} (about 37% of the original value) is called **skin depth** or **penetration depth** of the medium; that is,

$$E_0 e^{-\alpha \delta} = E_0 e^{-1}$$

or

$$\delta = \frac{1}{\alpha}$$

2. The **skin depth** is a measure of the depth to which an EM wave can penetrate the medium. $\delta = \frac{1}{\alpha}$ is generally valid for any material medium.
3. For good conductors,

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \frac{1}{\sqrt{\pi f\mu\sigma}} = \frac{1}{\alpha}$$

The illustration in Figure 10.8 for a good conductor is exaggerated. However, for a partially conducting medium, the skin depth can be quite large. For a good conductor,

$$\eta = \frac{1}{\sigma\delta} \sqrt{2} e^{j\pi/4} = \frac{1+j}{\sigma\delta}$$

Noting that for good conductors we have $\alpha = \beta = \frac{1}{\delta}$,

$$E = E_0 e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta}\right) \mathbf{a}_x$$

showing that δ measures the exponential damping of the wave as it travels through the conductor.

4. The phenomenon whereby field intensity in a conductor rapidly decreases is known as the **skin effect**. It is a tendency of charges to migrate from the bulk of the conducting material to the surface, resulting in higher resistance. The fields and associated currents are confined to a very thin layer (the skin) of the conductor surface. For a wire of radius a , for example, it is a good approximation at high frequencies to assume that all of the current flows in the circular ring of thickness δ .

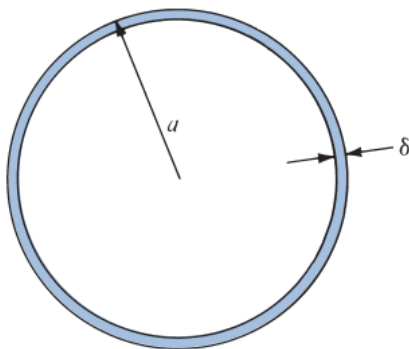


FIGURE 10.9 Skin depth at high frequencies, $\delta \ll a$.

5. The skin effect appears in different guises in such problems as attenuation in waveguides, effective or ac resistance of transmission lines, and electromagnetic shielding. It is used to advantage in many applications. For example, because the skin depth in silver is very small, the difference in performance between a pure silver component and a silver-plated brass component is negligible, so silver plating is often used to reduce the material cost of waveguide components. For the same reason, hollow tubular conductors are used instead of solid conductors in outdoor television antennas. Effective electromagnetic shielding of electrical devices can be provided by conductive enclosures a few skin depths in thickness.

TABLE 10.2 Skin Depth in Copper*

Frequency (Hz)	10	60	100	500	10^4	10^8	10^{10}
Skin depth (mm)	20.8	8.6	6.6	2.99	0.66	6.6×10^{-3}	6.6×10^{-4}

*For copper, $\sigma = 5.8 \times 10^7$ S/m, $\mu = \mu_0$, $\delta = 66.1/\sqrt{f}$ (in mm).

6. The skin depth is useful in calculating the *ac resistance* due to skin effect. The resistance given below is called the *dc resistance*, that is,

$$R_{dc} = \frac{\ell}{\sigma S}$$

We define the *surface or skin resistance* R_s (in Ω) as the real part of η for a good conductor. Thus

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\pi f \mu}{\sigma}}$$

This is the resistance of a unit width and unit length of the conductor. It is equivalent to the dc resistance for a unit length of the conductor having cross-sectional area $1 \times \delta$. Thus for a given width w and length ℓ , the ac resistance is calculated by using the familiar dc resistance relation of $R_{dc} = \frac{\ell}{\sigma S}$ and assuming a uniform current flow in the conductor of thickness δ ; that is,

$$R_{ac} = \frac{\ell}{\sigma \delta w} = \frac{R_s \ell}{w}$$

where $S \simeq \delta w$.

For a conductor wire of radius a (see Figure 10.9), $w = 2\pi a$, so

$$\frac{R_{ac}}{R_{dc}} = \frac{\frac{\ell}{\sigma 2\pi a \delta}}{\frac{\ell}{\sigma \pi a^2}} = \frac{a}{2\delta} = \frac{a}{2} \sqrt{\pi f \mu \sigma}$$

Since $\delta \ll a$ at high frequencies, this shows that R_{ac} is far greater than R_{dc} . In general, the ratio of the ac to the dc resistance starts at 1.0 for dc and very low frequencies and increases as the frequency increases. Also, although the bulk of the current is nonuniformly distributed over a thickness of 5δ of the conductor, the power loss is the same as though it were uniformly distributed over a thickness of δ and zero elsewhere. This is one more reason that δ is referred to as the skin depth.

2.1.5 Summary of the case study

For easy reference, the formulas for propagation constants for different media are summarized in Table 10.3.

TABLE 10.3 Formulas, for α , β , η , ν , and λ

	Lossy Medium	Lossless Medium	Free Space	Conductor
$\alpha =$	$\omega \left[\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right] \right]^{1/2}$	0	0	$\sqrt{\pi f \mu \sigma}$
$\beta =$	$\omega \left[\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu\epsilon}$	$\omega \sqrt{\mu_0\epsilon_0}$	$\sqrt{\pi f \mu \sigma}$
$\eta =$	$\sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377$	$(1 + j) \frac{\alpha}{\sigma}$
$\nu = \frac{\omega}{\beta}, \quad \lambda = \frac{2\pi}{\beta}$				

2.2 Plane wave incident on the boundary between media

2.2.1 A plane wave at normal incidence

1. Unlike uniform plane waves traveling in unbounded, homogeneous, isotropic media, when a plane wave from one medium meets a **different medium**, it is partly **reflected** and partly **transmitted**. The proportion of the incident wave that is reflected or transmitted depends on the **constitutive parameters** (ε, μ, σ) of the two media involved.
2. Suppose that a plane wave propagating along the $+z$ -direction is incident normally on the boundary $z = 0$ between medium 1 ($z < 0$) characterized by $\sigma_1, \varepsilon_1, \mu_1$ and medium 2 ($z > 0$) characterized by $\sigma_2, \varepsilon_2, \mu_2$, as shown in Figure 10.12. In Figure 10.12, subscripts i, r, t denote incident, reflected, and transmitted waves, respectively.

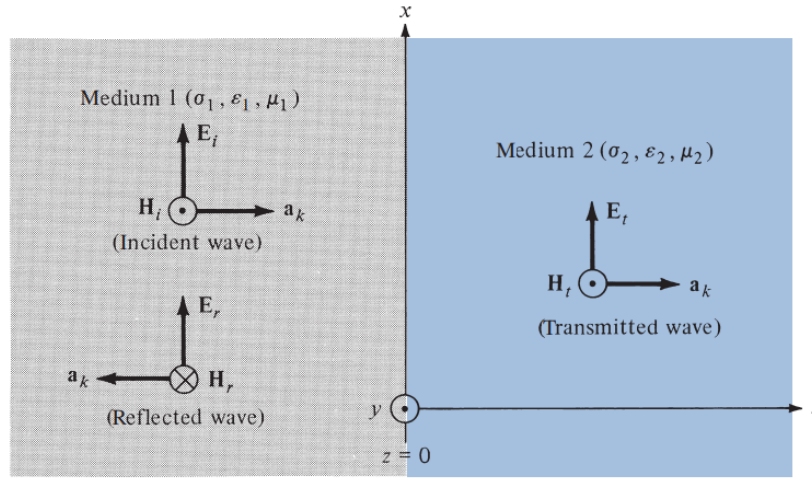


FIGURE 10.12 A plane wave incident normally on an interface between two different media.

- (a) **Incident wave** ($\mathbf{E}_i, \mathbf{H}_i$): medium 1, traveling along $+z$ direction, the phasor forms of fields are

$$\mathbf{E}_{is} = E_{i0}e^{-\gamma_1 z}\mathbf{a}_x \quad \mathbf{H}_{is} = \frac{E_{i0}}{\eta_1}e^{-\gamma_1 z}\mathbf{a}_y \equiv H_0e^{-\gamma_1 z}\mathbf{a}_y$$

- (b) **Reflected wave** ($\mathbf{E}_r, \mathbf{H}_r$): medium 1, traveling along $-z$ direction, the phasor forms of fields are

$$\mathbf{E}_{rs} = E_{r0}e^{+\gamma_1 z}\mathbf{a}_x, \quad \mathbf{H}_{rs} = \frac{E_{r0}}{\eta_1}e^{+\gamma_1 z}(-\mathbf{a}_y) \equiv -H_{r0}e^{+\gamma_1 z}\mathbf{a}_y$$

- (c) **Transmitted wave** ($\mathbf{E}_t, \mathbf{H}_t$): medium 2, traveling along $+z$ direction, the phasor forms of fields are

$$\mathbf{E}_{ts} = E_{t0}e^{-\gamma_2 z}\mathbf{a}_x, \quad \mathbf{H}_{ts} = \frac{E_{t0}}{\eta_2}e^{-\gamma_2 z}\mathbf{a}_y \equiv H_{t0}e^{-\gamma_2 z}\mathbf{a}_y$$

E_{i0}, E_{r0} , and E_{t0} are, respectively, the magnitudes of the incident, reflected, and transmitted electric fields at the interface ($z = 0$).

3. Recall the electromagnetic boundary conditions

$$\begin{aligned}
E_{1t} - E_{2t} &= 0 \quad \text{or} \quad (\mathbf{E}_1 - \mathbf{E}_2) \times \hat{n} = 0 \\
H_{1t} - H_{2t} &= K \quad \text{or} \quad (\mathbf{H}_1 - \mathbf{H}_2) \times \hat{n} = \mathbf{K} \\
D_{1n} - D_{2n} &= \rho_s \quad \text{or} \quad (\mathbf{D}_1 - \mathbf{D}_2) \cdot \hat{n} = \rho_s \\
B_{1n} - B_{2n} &= 0 \quad \text{or} \quad (\mathbf{B}_2 - \mathbf{B}_1) \cdot \hat{n} = 0
\end{aligned}$$

At the interface $z = 0$, we assume that there is no free charge ($\rho_s = 0$) or no free current ($K = 0$). Then the boundary conditions require that the **tangential** components of $\mathbf{E}$ and $\mathbf{H}$ fields must be continuous.

$$\begin{aligned}
E_{1t} - E_{2t} &= 0 \\
H_{1t} - H_{2t} &= 0
\end{aligned}$$

Since the waves are transverse, $\mathbf{E}$ and $\mathbf{H}$ fields are entirely tangential to the interface. Hence at $z = 0$, $\mathbf{E}_{1 \tan} = \mathbf{E}_{2 \tan}$ and $\mathbf{H}_{1 \tan} = \mathbf{H}_{2 \tan}$ imply that

$$\begin{aligned}
\mathbf{E}_i(0) + \mathbf{E}_r(0) &= \mathbf{E}_t(0) \quad \rightarrow \quad E_{i0} + E_{r0} = E_{t0} \\
\mathbf{H}_i(0) + \mathbf{H}_r(0) &= \mathbf{H}_t(0) \quad \rightarrow \quad \frac{1}{\eta_1} (E_{i0} - E_{r0}) = \frac{E_{t0}}{\eta_2}
\end{aligned}$$

4. Solving E_{r0}, E_{t0} in terms of E_{i0} gives

$$E_{r0} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} E_{i0} \equiv \Gamma E_{i0}$$

and

$$E_{t0} = \frac{2\eta_2}{\eta_2 + \eta_1} E_{i0} \equiv \tau E_{i0}$$

where we have defined the *reflection coefficient* Γ and the *transmission coefficient* τ as

$$\Gamma \equiv \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \tau \equiv \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Note that

- (a) $1 + \Gamma = \tau$
- (b) Both Γ and τ are dimensionless and may be complex.
- (c) $0 \leq |\Gamma| \leq 1$

Case study 1. Perfect dielectric and perfect conductor.

1. In this case, the media are characterized by

$$\sigma_1 \approx 0 \text{ (perfect dielectric),} \quad \sigma_2 \rightarrow \infty \text{ (perfect conductor)}$$

Then, the corresponding impedances are

$$\eta_1 = \sqrt{\frac{j\omega\mu_1}{\sigma_1 + j\omega\varepsilon_1}} = \sqrt{\frac{\mu_1}{\varepsilon_1}}, \quad \eta_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2 + j\omega\varepsilon_2}} = 0$$

and the reflection and transmission coefficients become

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -1, \quad \tau = 1 + \Gamma = 0$$

Meanwhile, the attenuation coefficients are

$$\alpha_1 = \omega \sqrt{\frac{\mu_1\varepsilon_1}{2} \left(\sqrt{1 + \left(\frac{\sigma_1}{\omega\varepsilon_1} \right)^2} - 1 \right)} = 0, \quad \alpha_2 = \omega \sqrt{\frac{\mu_2\varepsilon_2}{2} \left(\sqrt{1 + \left(\frac{\sigma_2}{\omega\varepsilon_2} \right)^2} - 1 \right)} = \infty$$

Then the skin depth of the medium 2 is given by

$$\delta_2 = \frac{1}{\alpha_2} = \frac{1}{\sqrt{\pi f \mu_2 \sigma_2}} = 0$$

Hence, there is no transmitted wave in medium 2, ($\mathbf{E}_2 = 0$) and the incident wave is totally reflected (with a phase difference $\Gamma = -1 = e^{j\pi}$).

2. The totally reflected wave interferes with the incident wave to form a *standing wave*, which does not travel. We can describe the formation of this standing wave in medium 1 as follows. Using the phasor form of incident and reflected waves in medium 1,

$$\begin{aligned} \mathbf{E}_{1s} &= \mathbf{E}_{is} + \mathbf{E}_{rs} = (E_{i0}e^{-\gamma_1 z} + E_{r0}e^{+\gamma_1 z}) \mathbf{a}_x \\ &= (E_{i0}e^{-j\beta_1 z} + \Gamma E_{i0}e^{+j\beta_1 z}) \mathbf{a}_x \\ &= -E_{i0} (e^{+j\beta_1 z} - e^{-j\beta_1 z}) \mathbf{a}_x \\ &= E_{i0} \cdot 2j \sin(\beta_1 z) \mathbf{a}_x \end{aligned}$$

Then the instantaneous field is

$$\mathbf{E}_1 = \text{Re}(\mathbf{E}_{1s}e^{j\omega t}) = 2E_{i0} \sin(\beta_1 z) \sin(\omega t) \mathbf{a}_x$$

Similarly,

$$\begin{aligned} \mathbf{H}_{1s} &= \mathbf{H}_{is} + \mathbf{H}_{rs} = H_{i0}e^{-\gamma_1 z} \mathbf{a}_y + H_{r0}e^{+\gamma_1 z} (-\mathbf{a}_y) \\ &= \left(\frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} - \Gamma \frac{E_{i0}}{\eta_1} e^{+j\beta_1 z} \right) \mathbf{a}_y \\ &= \frac{E_{i0}}{\eta_1} (e^{+j\beta_1 z} + e^{-j\beta_1 z}) \mathbf{a}_y \\ &= \frac{2E_{i0}}{\eta_1} \cdot 2 \cos(\beta_1 z) \mathbf{a}_y \\ \therefore \mathbf{H}_1 &= \text{Re}(\mathbf{H}_{1s}e^{j\omega t}) = \frac{2E_{i0}}{\eta_1} \cos(\beta_1 z) \cos(\omega t) \mathbf{a}_y \end{aligned}$$

These results are illustrated in the following figure:

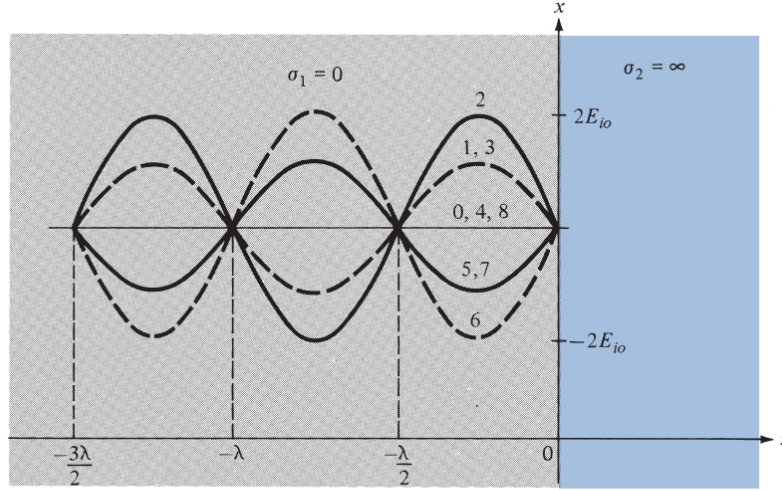


FIGURE 10.13 Standing waves $E = 2E_{i0} \sin \beta_1 z \sin \omega t \mathbf{a}_x$. The curves 0, 1, 2, 3, 4, ..., are, respectively, at times $t = 0, T/8, T/4, 3T/8, T/2, \dots$; $\lambda = 2\pi/\beta_1$.

Case study 2. Two perfect dielectrics

1. In this case, the media are characterized by

$$\sigma_1 \approx \sigma_2 \approx 0 \text{ (perfect dielectric)}$$

Then, the corresponding impedances are

$$\eta_1 = \sqrt{\frac{j\omega\mu_1}{\sigma_1 + j\omega\varepsilon_1}} = \sqrt{\frac{\mu_1}{\varepsilon_1}}, \quad \eta_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2 + j\omega\varepsilon_2}} = \sqrt{\frac{\mu_2}{\varepsilon_2}}$$

and the reflection and transmission coefficients become nonzero values:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \neq 0, \quad \tau = 1 + \Gamma \neq 0$$

Meanwhile, the attenuation coefficients are

$$\alpha_1 = \omega \sqrt{\frac{\mu_1 \varepsilon_1}{2} \left(\sqrt{1 + \left(\frac{\sigma_1}{\omega \varepsilon_1} \right)^2} - 1 \right)} = 0, \quad \alpha_2 = \omega \sqrt{\frac{\mu_2 \varepsilon_2}{2} \left(\sqrt{1 + \left(\frac{\sigma_2}{\omega \varepsilon_2} \right)^2} - 1 \right)} = 0$$

Hence, there is a partially transmitted wave in medium 2 ($\mathbf{E}_2 \neq 0$), and the incident wave is partially reflected.

2. Even in this case, the partially reflected wave interferes with the incident wave to form a *standing wave*. However, the incident and reflected waves have amplitudes that are not equal in magnitude. We can obtain the envelope $|\mathbf{E}_1|$ of the standing wave in medium 1 as follows. Using the phasor form of incident and reflected waves in medium 1,

$$\begin{aligned} \mathbf{E}_{1s} &= \mathbf{E}_{is} + \mathbf{E}_{rs} = (E_{i0}e^{-\gamma_1 z} + E_{r0}e^{+\gamma_1 z}) \mathbf{a}_x \\ &= (E_{i0}e^{-j\beta_1 z} + \Gamma E_{i0}e^{+j\beta_1 z}) \mathbf{a}_x \\ &= E_{i0} (e^{-j\beta_1 z} + \Gamma e^{+j\beta_1 z}) \mathbf{a}_x \end{aligned}$$

Hence, the field envelope $|\mathbf{E}_1|$ is obtained by the complex square of the phasor form, $\mathbf{E}_{1s}$,

$$\begin{aligned} |\mathbf{E}_1| &= \sqrt{|\mathbf{E}_{1s}|^2} = \sqrt{E_{i0} (e^{-j\beta_1 z} + \Gamma e^{+j\beta_1 z}) \times E_{i0} (e^{+j\beta_1 z} + \Gamma e^{-j\beta_1 z})} \\ &= E_{i0} \sqrt{1 + \Gamma^2 + 2\Gamma \cos(2\beta_1 z)} \end{aligned}$$

- (a) If $\eta_2 > \eta_1$, $\Gamma > 0$ and relative maximums of $|\mathbf{E}_1|$ occur at $-2\beta_1 z = 2n\pi$ ($n = 0, 1, 2, \dots$), the minimums occur at $-2\beta_1 z = (2n + 1)\pi$.
- (b) If $\eta_2 < \eta_1$, $\Gamma < 0$, and relative minimums of $|\mathbf{E}_1|$ occur at $-2\beta_1 z = 2n\pi$ ($n = 0, 1, 2, \dots$) and the maximums occur at $-2\beta_1 z = (2n + 1)\pi$.

Similarly,

$$\begin{aligned} \mathbf{H}_{1s} &= \mathbf{H}_{is} + \mathbf{H}_{rs} = H_{i0} e^{-\gamma_1 z} \mathbf{a}_y + H_{r0} e^{+\gamma_1 z} (-\mathbf{a}_y) \\ &= \left(\frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} - \Gamma \frac{E_{i0}}{\eta_1} e^{+j\beta_1 z} \right) \mathbf{a}_y \\ &= \frac{E_{i0}}{\eta_1} (e^{-j\beta_1 z} - \Gamma e^{+j\beta_1 z}) \mathbf{a}_y \\ \therefore |\mathbf{H}_{1s}| &= \sqrt{\frac{E_{i0}}{\eta_1} (e^{-j\beta_1 z} - \Gamma e^{+j\beta_1 z}) \times \frac{E_{i0}}{\eta_1} (e^{+j\beta_1 z} - \Gamma e^{-j\beta_1 z})} \\ &= \frac{E_{i0}}{\eta_1} \sqrt{1 + \Gamma^2 - 2\Gamma \cos(2\beta_1 z)} \end{aligned}$$

Thus, $|\mathbf{H}_1|$ minimum occurs whenever there is $|\mathbf{E}_1|$ maximum, and vice versa. The transmitted wave in medium 2 is a purely traveling wave, and consequently there are no maxima and minima in this region.

3. This standing wave is illustrated in the following figure:

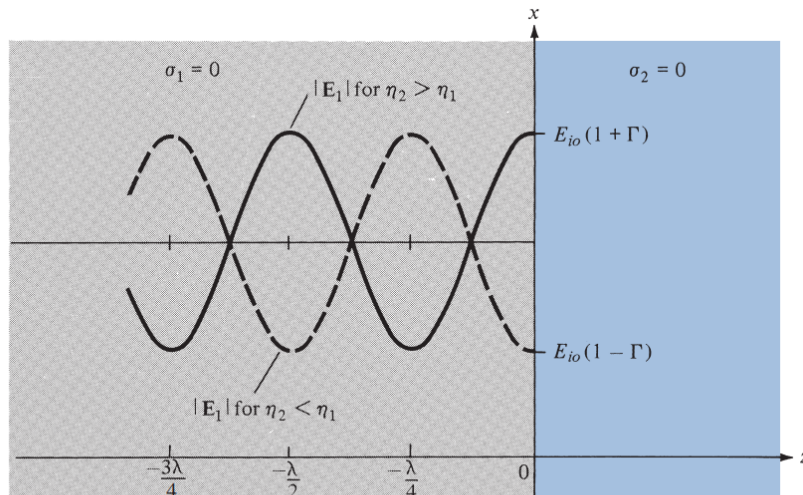


FIGURE 10.14 Standing waves due to reflection at an interface between two lossless media; $\lambda = 2\pi/\beta_1$.

We define the degree of standing wave formation as *the standing wave ratio* s , which is defined as

$$s = \frac{|E_1|_{\max}}{|E_1|_{\min}} = \frac{|H_1|_{\max}}{|H_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

If we measure the standing wave ratio, we can obtain the magnitude of the reflection coefficient by

$$|\Gamma| = \frac{s - 1}{s + 1}$$

Since $|\Gamma| \leq 1$, it follows that $1 \leq s \leq \infty$. The standing wave ratio is dimensionless, and it is customarily expressed in decibels (dB) as

$$s_{\text{dB}} = 20 \log_{10} s$$

2.2.2 A plane wave at oblique incidence

Preliminary A. Maxwell equations of a uniform plane wave

1. Assume a **lossless** ($\sigma_1 = \sigma_2 = 0$) media. Then, the corresponding impedances are

$$\eta_1 = \sqrt{\frac{j\omega\mu_1}{\sigma_1 + j\omega\varepsilon_1}} = \sqrt{\frac{\mu_1}{\varepsilon_1}}, \quad \eta_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2 + j\omega\varepsilon_2}} = \sqrt{\frac{\mu_2}{\varepsilon_2}}$$

A uniform plane wave takes the general form of

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) = \text{Re}[\mathbf{E}_0 e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)}] \\ \therefore \mathbf{E}_s(\mathbf{r}) &= \mathbf{E}_0 e^{j(\mathbf{k} \cdot \mathbf{r})} \end{aligned}$$

where $\mathbf{E}_0$ is a constant vector.

2. Let $\mathbf{r} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z$ be a position vector and $\mathbf{k} = k_x\mathbf{a}_x + k_y\mathbf{a}_y + k_z\mathbf{a}_z$ be a *propagation vector*. The magnitude of $\mathbf{k}$ is related to ω according to the *dispersion* relation

$$\mathbf{k}^2 = k_x^2 + k_y^2 + k_z^2 = \omega^2 \mu \varepsilon$$

Here, dispersion refers to the phenomenon of signal distortion due to a dependence of the phase velocity on frequency.

3. We can reduce the phasor form of Maxwell's equations for a uniform plane wave in a *source-free* region.

$$\begin{aligned} \nabla \times \mathbf{E}_s &= j\omega\mu\mathbf{H}_s \\ \nabla \times \mathbf{E}_s &= \nabla \times (\mathbf{E}_0 e^{j\mathbf{k} \cdot \mathbf{r}}) \\ &= \frac{1}{h_1 h_2 h_3} \epsilon_{ijk} (h_i \hat{e}_i) \partial_j (E_{0k} \cdot e^{j\mathbf{k} \cdot \mathbf{r}}) \\ &= \frac{1}{h_1 h_2 h_3} \epsilon_{ijk} (h_i \hat{e}_i) (\partial_j (E_{0k}) e^{j\mathbf{k} \cdot \mathbf{r}} + E_{0k} \partial_j (e^{j\mathbf{k} \cdot \mathbf{r}})) \\ &= \frac{1}{h_1 h_2 h_3} \epsilon_{ijk} (h_i \hat{e}_i) (0 + E_{0k} (jk_j) e^{j\mathbf{k} \cdot \mathbf{r}}) \\ &= \frac{1}{h_1 h_2 h_3} \epsilon_{ijk} (h_i \hat{e}_i) [(jk_j) E_{0k} e^{j\mathbf{k} \cdot \mathbf{r}}] = \frac{1}{h_1 h_2 h_3} \epsilon_{ijk} (h_i \hat{e}_i) [(jk_j) \mathbf{E}_{sk}] \\ &= (j\mathbf{k}) \times \mathbf{E}_s \\ \therefore \mathbf{k} \times \mathbf{E}_s &= \omega\mu\mathbf{H}_s \end{aligned}$$

Similarly,

$$\mathbf{k} \times \mathbf{H}_s = -\omega\varepsilon\mathbf{E}$$

Also,

$$\begin{aligned} 0 &= \nabla \cdot \mathbf{E}_s = \partial_i (\mathbf{E}_{si}) = \partial_i (\mathbf{E}_{0i} e^{j\mathbf{k} \cdot \mathbf{r}}) \\ &= \mathbf{E}_{0i} \partial_i (e^{j\mathbf{k} \cdot \mathbf{r}}) = \mathbf{E}_{0i} (jk_i) (e^{j\mathbf{k} \cdot \mathbf{r}}) \\ &= (jk_i) (\mathbf{E}_{0i} e^{j\mathbf{k} \cdot \mathbf{r}}) = j\mathbf{k} \cdot \mathbf{E}_s \\ \therefore \mathbf{k} \cdot \mathbf{E}_s &= 0 \end{aligned}$$

Likewise,

$$\mathbf{k} \cdot \mathbf{H}_s = 0$$

4. To summarize,

$$\mathbf{k} \times \mathbf{E}_s = \omega\mu\mathbf{H}_s, \quad \mathbf{k} \times \mathbf{H}_s = -\omega\varepsilon\mathbf{E}_s, \quad \mathbf{k} \cdot \mathbf{E}_s = 0, \quad \mathbf{k} \cdot \mathbf{H}_s = 0$$

This results shows that

- (a) $\mathbf{E}$, $\mathbf{H}$ and $\mathbf{k}$ are mutually orthogonall
- (b) $\mathbf{E}$, $\mathbf{H}$ lie on the plane

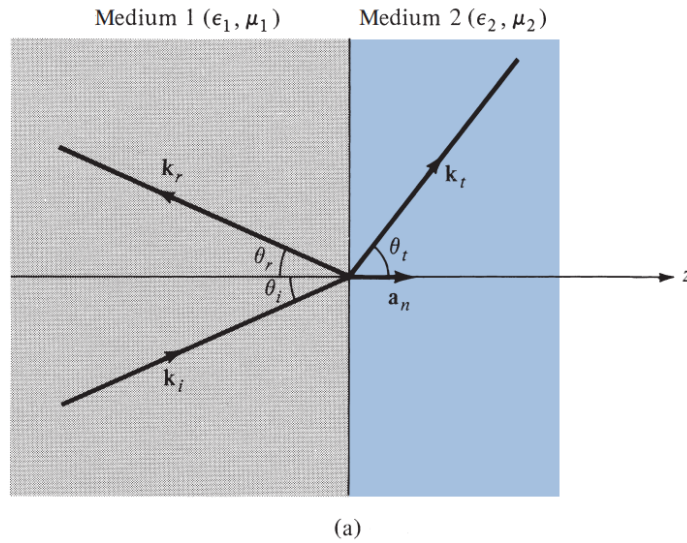
$$\mathbf{k} \cdot \mathbf{r} = k_x x + k_y y + k_z z = \text{const.}$$

and the $\mathbf{H}$ field corresponding to the given $\mathbf{E}$ field can be obtained by

$$\mathbf{H} = \frac{1}{\omega\mu} \mathbf{k} \times \mathbf{E} = \frac{k}{\omega\mu} \mathbf{a}_k \times \mathbf{E} = \frac{1}{\eta} \mathbf{a}_k \times \mathbf{E}$$

Preliminary B. Phase matching conditions

1. Let us consider the oblique incidence of a uniform plane wave at a plane boundary as illustrated in the following figure:



The *plane of incidence* is defined by the $\mathbf{k}$ and a unit normal vector $\mathbf{a}_n$ to the boundary. Here, the angle θ_i between $\mathbf{k}$ and $\mathbf{a}_n$ is the *angle of incidence*.

2. We can write the field expressions for the incident, reflected and transmitted wave as

$$\mathbf{E}_i = \mathbf{E}_{i0} \cos(k_{ix}x + k_{iy}y + k_{iz}z - \omega_i t)$$

$$\mathbf{E}_r = \mathbf{E}_{r0} \cos(k_{rx}x + k_{ry}y + k_{rz}z - \omega_r t)$$

$$\mathbf{E}_t = \mathbf{E}_{t0} \cos(k_{tx}x + k_{ty}y + k_{tz}z - \omega_t t)$$

3. Since the **tangential component** of $\mathbf{E}$ must be continuous across the boundary $z = 0$,

$$\mathbf{E}_i(z = 0) + \mathbf{E}_r(z = 0) = \mathbf{E}_t(z = 0)$$

This boundary conditions can be satisfied for all x, y only if

1. $\omega_i = \omega_r = \omega_t = \omega$: The frequency is unchanged
2. $k_{ix} = k_{rx} = k_{tx} = k_x$: tangential components of the propagation vectors be continuous
3. $k_{iy} = k_{ry} = k_{ty} = k_y$: tangential components of the propagation vectors be continuous

In other words, This means that the propagation vectors $\mathbf{k}_i, \mathbf{k}_t$, and $\mathbf{k}_r$ must all lie in the plane of incidence. Thus, by conditions 2 and 3,

$$\begin{aligned} k_i \sin \theta_i &= k_r \sin \theta_r \\ k_i \sin \theta_i &= k_t \sin \theta_t \end{aligned}$$

where θ_r is *the angle of reflection* and θ_t is *the angle of transmission*. But for lossless media,

$$\begin{aligned} k_i &= k_r = \omega \sqrt{\mu_1 \varepsilon_1} \\ k_t &= \omega \sqrt{\mu_2 \varepsilon_2} \end{aligned}$$

It is clear that from $k_i = k_r$, we retrieve the law of reflection

$$\boxed{\theta_r = \theta_i}$$

On the other hand, the law of refraction, *Snell's law*, is also clear from

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_i}{k_t} = \frac{u_2}{u_1} = \sqrt{\frac{\mu_1 \varepsilon_1}{\mu_2 \varepsilon_2}}$$

where $u = \omega/k$ is the phase velocity.

$$\boxed{n_1 \sin \theta_i = n_2 \sin \theta_t}$$

where $n_1 = c\sqrt{\mu_1 \varepsilon_1} = c/u_1$ and $n_2 = c\sqrt{\mu_2 \varepsilon_2} = c/u_2$ are the *refractive indices* of the media.

Case study 1. Parallel polarization : Fresnel equations and Brewster effect

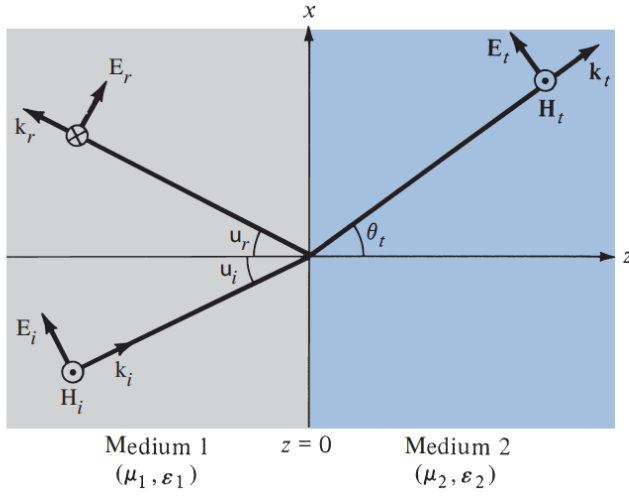


FIGURE 10.17 Oblique incidence with $\mathbf{E}$ parallel to the plane of incidence.

1. In Figure 10.17, the propagation vectors can be written as

$$\begin{aligned}\mathbf{k}_i &= k_i(\cos \theta_i \mathbf{a}_z + \sin \theta_i \mathbf{a}_x) \\ \mathbf{k}_r &= k_r(-\cos \theta_i \mathbf{a}_z + \sin \theta_i \mathbf{a}_x) \\ \mathbf{k}_t &= k_t(\cos \theta_t \mathbf{a}_z + \sin \theta_t \mathbf{a}_x)\end{aligned}$$

where $k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $k_2 = \omega\sqrt{\mu_2\epsilon_2}$. Now the electric fields of each wave can be found by imposing $\mathbf{k} \cdot \mathbf{E}_s = 0$,

$$\begin{aligned}\mathbf{E}_{is} &= (E_{iz}\mathbf{a}_z + E_{ix}\mathbf{a}_x)e^{i\mathbf{k}_i \cdot \mathbf{r}} = E_{i0}(-\sin \theta_i \mathbf{a}_z + \cos \theta_i \mathbf{a}_x)e^{-jk_1(z \cos \theta_i + x \sin \theta_i)} \\ \mathbf{E}_{rs} &= (E_{rz}\mathbf{a}_z + E_{rx}\mathbf{a}_x)e^{i\mathbf{k}_r \cdot \mathbf{r}} = E_{r0}(\sin \theta_r \mathbf{a}_z + \cos \theta_r \mathbf{a}_x)e^{+jk_1(z \cos \theta_r - x \sin \theta_r)} \\ \mathbf{E}_{ts} &= (E_{tz}\mathbf{a}_z + E_{tx}\mathbf{a}_x)e^{i\mathbf{k}_t \cdot \mathbf{r}} = E_{t0}(-\sin \theta_t \mathbf{a}_z + \cos \theta_t \mathbf{a}_x)e^{-jk_2(z \cos \theta_t + x \sin \theta_t)}\end{aligned}$$

and the $\mathbf{H}_s$ can be found by $\mathbf{H}_s = \mathbf{a}_k \times \frac{\mathbf{E}_s}{\eta}$

$$\begin{aligned}\mathbf{H}_{is} &= \mathbf{a}_{ki} \times \frac{\mathbf{E}_{is}}{\eta_1} = \frac{E_{i0}}{\eta_1} e^{-jk_1(z \cos \theta_i + x \sin \theta_i)} \mathbf{a}_y \\ \mathbf{H}_{rs} &= \mathbf{a}_{kr} \times \frac{\mathbf{E}_{rs}}{\eta_1} = \frac{E_{r0}}{\eta_1} e^{+jk_1(z \cos \theta_r - x \sin \theta_r)} (-\mathbf{a}_y) \\ \mathbf{H}_{ts} &= \mathbf{a}_{kt} \times \frac{\mathbf{E}_{ts}}{\eta_2} = \frac{E_{t0}}{\eta_2} e^{-jk_2(z \cos \theta_t + x \sin \theta_t)} \mathbf{a}_y\end{aligned}$$

2. Now we impose the law of reflection ($\theta_r = \theta_i$) and the continuity of tangial components at the boundary $z = 0$,

$$\mathbf{E}_{is}(z=0) + \mathbf{E}_{rs}(z=0) = \mathbf{E}_{ts}(z=0)$$

$$\begin{aligned}E_{i0}(-\sin \theta_i \mathbf{a}_z + \cos \theta_i \mathbf{a}_x)e^{-jk_1 x \sin \theta_i} + E_{r0}(\sin \theta_i \mathbf{a}_z + \cos \theta_i \mathbf{a}_x)e^{-jk_1 x \sin \theta_i} \\ = E_{t0}(-\sin \theta_t \mathbf{a}_z + \cos \theta_t \mathbf{a}_x)e^{-jk_2 x \sin \theta_t}\end{aligned}$$

Next, we employ the phase matching condition $k_i \sin \theta_i = k_t \sin \theta_t$ to eliminate the exponential factors at each terms,

$$\begin{aligned} E_{i0}(-\sin \theta_i \mathbf{a}_z + \cos \theta_i \mathbf{a}_x) + E_{r0}(\sin \theta_i \mathbf{a}_z + \cos \theta_i \mathbf{a}_x) &= E_{t0}(-\sin \theta_t \mathbf{a}_z + \cos \theta_t \mathbf{a}_x) \\ \Rightarrow \mathbf{a}_z [(-E_{i0} + E_{r0}) \sin \theta_i] + \mathbf{a}_x [(E_{i0} + E_{r0}) \cos \theta_i] &= -E_{t0} \sin \theta_t \mathbf{a}_z + E_{t0} \cos \theta_t \mathbf{a}_x \end{aligned}$$

Hence, we obtain

$$(-E_{i0} + E_{r0}) \sin \theta_i = -E_{t0} \sin \theta_t \quad (1)$$

$$(E_{i0} + E_{r0}) \cos \theta_i = E_{t0} \cos \theta_t \quad (2)$$

Similarly, the continuity of $\mathbf{H}_s$ at the boundary $z = 0$ yields

$$\frac{E_{i0} - E_{r0}}{\eta_1} = \frac{E_{t0}}{\eta_2} \quad (3)$$

Expressing E_{r0} and E_{t0} in terms of E_{i0} , we obtain

$$\boxed{\Gamma_{\parallel} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}}$$

or

$$E_{r0} = \Gamma_{\parallel} E_{i0}$$

and

$$\boxed{\tau_{\parallel} = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}}$$

or

$$E_{t0} = \tau_{\parallel} E_{i0}$$

These expressions for reflection and transmission coefficients are called *Fresnel equations*. $\Gamma_{\parallel}$ and $\tau_{\parallel}$ are called *Fresnel coefficient*.

3. **Limiting case study:** When $\theta_i \rightarrow 0$ (normal incidence), Snell's law assures $\theta_t \rightarrow 0$ and the Fresnel coefficients become

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} \rightarrow \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} \rightarrow \frac{2\eta_2}{\eta_2 + \eta_1}$$

4. The relationship between Fresnel coefficients can be obtained by application of Snell's law,

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \sqrt{1 - \left(\frac{u_2}{u_1}\right)^2 \sin^2 \theta_i}$$

it is easily shown that

$$1 + \Gamma_{\parallel} = \tau_{\parallel} \left(\frac{\cos \theta_t}{\cos \theta_i} \right)$$

5. **Brewster effect:** it is evident that it is possible that $\Gamma_{\parallel} = 0$ because the numerator is the difference of two terms. Under this condition, there is no reflection ($E_{r0} = 0$), and the incident angle at which this takes place is called the *Brewster angle* $\theta_{B\parallel}$.

The Brewster angle is obtained by setting $\theta_i = \theta_{B\parallel}$ when $\Gamma_{\parallel} = 0$,

$$\eta_2 \cos \theta_t = \eta_1 \cos \theta_{B\parallel} \quad \text{or} \quad \eta_2^2(1 - \sin^2 \theta_t) = \eta_1^2(1 - \sin^2 \theta_{B\parallel})$$

$$\boxed{\sin^2 \theta_{B\parallel} = \frac{1 - \mu_2 \varepsilon_1 / \mu_1 \varepsilon_2}{1 - (\varepsilon_1 / \varepsilon_2)^2}}$$

The Brewster angle is also known as the polarizing angle because an arbitrarily polarized incident wave will be reflected with only the component of $\mathbf{E}$ perpendicular to the plane of incidence. The Brewster effect is utilized in a laser tube where quartz windows are set at the Brewster angle to control polarization of emitted light.

6. It is of practical value to consider the case when the dielectric media are not only lossless but **nonmagnetic** as well—that is, $\mu_1 = \mu_2 = \mu_0$. For this situation,

$$\sin^2 \theta_{B\parallel} = \frac{1}{1 + \varepsilon_1 / \varepsilon_2} \quad \Rightarrow \quad \sin \theta_{B\parallel} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2}}$$

$$\text{or} \quad \tan \theta_{B\parallel} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = \frac{n_2}{n_1}$$

showing that there is a Brewster angle for any combination of ε_1 and ε_2 .

Case study 2. Perpendicular polarization

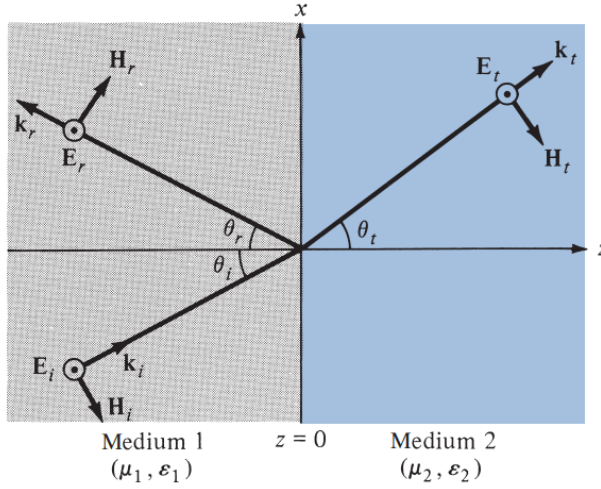


FIGURE 10.18 Oblique incidence with $\mathbf{E}$ perpendicular to the plane of incidence.

1. If $\mathbf{E}$ is perpendicularly polarized, $\mathbf{H}$ is parallel to the plane of incidence as depicted in Figure 10.18. Then the field expressions are

$$\begin{aligned}
 \mathbf{E}_{is} &= E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \mathbf{a}_y \\
 \mathbf{H}_{is} &= \frac{E_{i0}}{\eta_1} (-\cos \theta_i \mathbf{a}_x + \sin \theta_i \mathbf{a}_z) e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \\
 \mathbf{E}_{rs} &= E_{r0} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)} \mathbf{a}_y \\
 \mathbf{H}_{rs} &= \frac{E_{r0}}{\eta_1} (\cos \theta_r \mathbf{a}_x + \sin \theta_r \mathbf{a}_z) e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)} \\
 \mathbf{E}_{ts} &= E_{t0} e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)} \mathbf{a}_y \\
 \mathbf{H}_{ts} &= \frac{E_{t0}}{\eta_2} (-\cos \theta_t \mathbf{a}_x + \sin \theta_t \mathbf{a}_z) e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)}
 \end{aligned}$$

2. The phase matching conditions give

$$\begin{aligned}
 E_{i0} + E_{r0} &= E_{t0} \\
 \frac{1}{\eta_1} (E_{i0} - E_{r0}) \cos \theta_i &= \frac{1}{\eta_2} E_{t0} \cos \theta_t
 \end{aligned}$$

and the resulting Fresnel coefficients for perpendicular polarization are

$$\begin{aligned}
 \Gamma_{\perp} &= \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \\
 E_{r0} &= \Gamma_{\perp} E_{i0} \\
 \tau_{\perp} &= \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \\
 E_{t0} &= \tau_{\perp} E_{i0}
 \end{aligned}$$

with the relationship

$$1 + \Gamma_{\perp} = \tau_{\perp}$$

3. The Brewster angle for perpendicularly polarized case can be obtained in the similar manner:

$$\begin{aligned}\eta_2 \cos \theta_{B\perp} &= \eta_1 \cos \theta_t \\ \eta_2^2(1 - \sin^2 \theta_{B\perp}) &= \eta_1^2(1 - \sin^2 \theta_t)\end{aligned}$$

$$\sin^2 \theta_{B\perp} = \frac{1 - \mu_1 \varepsilon_2 / (\mu_2 \varepsilon_1)}{1 - (\mu_1 / \mu_2)^2}$$

For a lossless, nonmagnetic medium,

$$\begin{aligned}\sin \theta_{B\perp} &= \sqrt{\frac{\mu_2}{\mu_1 + \mu_2}} \\ \tan \theta_{B\perp} &= \sqrt{\frac{\mu_2}{\mu_1}}\end{aligned}$$